

ANSWERS & EXPLANATIONS
GENERAL STUDIES (P) TEST – 6336

- Q 1. C** • Linlithgow Statement (October 17, 1939): Following the outbreak of World War II, Lord Linlithgow issued a statement to gain Congress's support, which included setting up a consultative committee and discussing modifications to the Government of India Act of 1935.
- Muslim League's Lahore Resolution (March 23, 1940): Also known as the Pakistan Resolution, it was formally adopted by the Muslim League at its general session in Lahore, demanding independent states for Muslims in the North-West and East.
 - August Offer (August 8, 1940): Announced by Viceroy Lord Linlithgow at Simla, it proposed the expansion of the Executive Council and a constitution-making body after the war in exchange for Indian cooperation in World War II.
 - Cripps Mission (March 1942): Led by Sir Stafford Cripps, this mission was sent to India to seek support for the war effort by offering post-war dominion status and a constituent assembly.
 - Desai-Liaquat Pact (January 1945): A draft proposal for an interim government at the centre, **negotiated between Bhulabhai Desai (Congress) and Liaquat Ali Khan (Muslim League).**

Hence, option (c) is the correct answer.

- Q 2. B** • **Article 263 of the Constitution of India provides for the establishment of an Inter-State Council. It can set up the President based on the provisions of Article 263. Article 263 states:** If at any time the President deems it necessary to do so by forming a Council in the public interest:
- Investigate and advise on potential disputes between States;
 - Or to inquire into and discuss matters of common interest to some or all States or Union to one or more States; Or
 - Make recommendations on any such issue, especially recommendations for better coordination of policy and action on that issue,
- **It would be lawful for the President to establish such a Council and to define the nature of the duties to be performed by it and its organization and procedure. Hence, option (b) is the correct answer.**
- **Members of the Inter-state Council:**
- The Prime Minister is the Chairman of the Inter-State Council. The Council members are the Chief Ministers of all the States and Union Territories with Assembly, the Governors of the Union Territories without Assembly, the Governors of the States under the rule of the President, and the six Ministers in the Cabinet nominated by the Council Chairman. The five ministers nominated by the council chairman are regular invitees to the cabinet.

- Q 3. D** • **The Leader of the Opposition is the Leader of the largest opposition party.** The post was not recognised till 1969 when Ram Subhag Singh became the first recognised LoP of the Lok Sabha. The

office of the leader of the opposition is not mentioned in the Constitution. **The leader of Opposition in both the Houses were accorded statutory recognition under the Salaries and Allowances of Leader of Opposition in Parliament Act, 1977 and are entitled to the salary, allowances and other facilities equivalent to that of a cabinet minister. Hence, statement 2 is not correct.**

- **The Act defines the term 'Leader of the Opposition' as that member of the Lok Sabha or the Rajya Sabha who, for the time being, is the Leader of that House of the party in Opposition to the government having the greatest numerical strength and recognised, as such, by the chairman of the Rajya Sabha or the speaker of the Lok Sabha. However, no such 10 percent rule for the appointment of the Leader of Opposition is mentioned in the Leaders of Opposition in Parliament Act. Hence, statement 1 is not correct.**

Q 4. D • The FATF was formed during the 1989 G7 Summit in Paris to address the growing problem of money laundering.

- While the FATF is an independent body, it is housed at the OECD (Organisation for Economic Co-operation and Development) headquarters in France.
- India became a full member of the Financial Action Task Force (FATF) in June 2010.
- **Grey List Status:** India has never been placed on the FATF 'Grey List' (officially known as "Jurisdictions under Increased Monitoring").
- The FATF is an independent inter-governmental body, **not a specialized agency of the United Nations**. While it collaborates with the UN, it operates under its own mandate and does not have the power to 'enforce' international law in a judicial sense; instead, it sets international standards and uses peer reviews and public listings to encourage compliance. **Hence, option (d) is the correct answer.**

Q 5. A • The Aihole Inscription, composed by the Jain poet Ravikirti in 634–635 CE, is a 19-line Sanskrit eulogy on the Meguti Jain Temple in Aihole, Karnataka, celebrating the Chalukya king Pulakeshin II's victories, notably over Harshavardhana, and showcases Ravikirti's literary skill by comparing himself to Kalidasa and Bharavi, making it a key source for early Deccan history and literature. **Hence, Pair 1 is correctly matched.**

- The Prayag Prashasti (Allahabad Pillar Inscription) was written by Harisena, the court poet and minister of the Gupta Emperor Samudragupta. This Sanskrit inscription, found on an Ashokan pillar in Prayagraj, details Samudragupta's military victories, administration, and patronage of arts, making it a crucial historical source for the Gupta period. **Hence, Pair 2 is not correctly matched.**
- The famous Uttaramerur inscriptions (circa 919 CE and 921 CE) from the reign of Chola king Parantaka I, detailing local self-governance and the kuda-olai (pot-ticket) system, were not composed by a single known poet but rather recorded the resolutions of the Sabha (village assembly) of Uttaramerur. The records were inscribed on the walls of the Vaikunda Perumal Temple. **Hence, Pair 3 is not correctly matched.**

Q 6. C • The Treaty of Salbai, signed on May 17, 1782, formally concluded the First Anglo-Maratha War. It was negotiated between the British East India Company, represented by Warren Hastings, and the Maratha Empire, represented by Mahadji Shinde.

- It ushered in a period of relative peace between the Marathas and the British East India Company that lasted for nearly 20 years until the Second Anglo-Maratha War in 1802. **Hence, statement 1 is**

correct.

- One of the key strategic terms of the treaty was that the Marathas agreed to assist the British against Hyder Ali of Mysore. Specifically, the Marathas pledged to defeat Hyder Ali and reclaim their lost territories in the Carnatic, while the British agreed to recognise Madhavrao II as the legitimate Peshwa. **Hence, statement 2 is correct.**

Q 7. A • Under Article 142 of the Constitution, the Supreme Court has been vested with power to punish for contempt of Court including the power to punish for contempt of itself. In India, contempt is classified under two major categories:

- **Civil contempt:** According to section 2(b) of the Contempt of Courts Act, 1971 civil contempt means willful disobedience to any judgement, decree, direction, order, writ or other process of a court or willful breach of an undertaking given to a court.
- **Criminal contempt:** According to section 2(c) of The Contempt of Courts Act, 1971, criminal contempt means the publication (whether by word, spoken or written, or by signs, or by visible representation, or otherwise) of any matter or the doing of any other act whatsoever which:
 - Scandalizes or tends to scandalize, or lowers or tends to lower the authority of, any court, or
 - Prejudices or interferes or tends to interfere with the due course of any judicial proceeding, or
 - Interferes or tends to interfere with, or obstruct or tends to obstruct, the administration of justice in any other manner. **Hence, statement 1 is correct.**

- **Section 20 deals with period of limitation for initiating contempt proceeding. Section 20 provides that no court shall initiate contempt proceedings either on its own notions or otherwise after the expiry of one year from the date on which contempt is alleged to have been committed.** The period of limitation is applicable both in civil as well as criminal contempt. Contempt proceedings can be initiated either by filing an application or by the court itself suo moto. In both the cases, contempt proceedings must be initiated within one year from the date on which contempt is alleged to have been committed. **Hence, statement 2 is correct.**

Q 8. A • Article 191 of the Indian Constitution deals with the disqualifications for membership, it states that, a person shall be disqualified for being chosen as, and for being, a member of the Legislative Assembly or Legislative Council of a State-

(a) if he holds any office of profit under the Government of India or the Government of any State specified in the First Schedule, other than an office declared by the Legislature of the State by law not to disqualify its holder:

(b) if he is of unsound mind and stands so declared by a competent court;

(c) if he is an undischarged insolvent;

(d) if he is not a citizen of India, or has voluntarily acquired the citizenship of a foreign State, or is under any acknowledgment of allegiance or adherence to a foreign State;

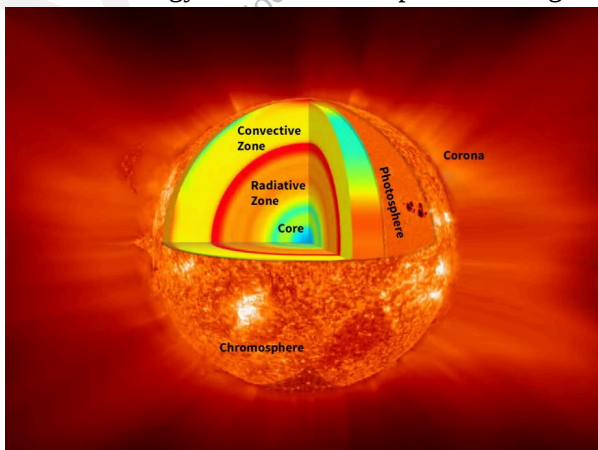
(e) if he is so disqualified by or under any law made by Parliament. **Hence, statement 1 is correct.**

- **The 74th Constitutional Amendment Act, 1992 (74th CAA) came into effect** on 1 June 1993, introduced Part IX A (the Municipalities) which deals with the issues relating to municipalities, in an article 243 P to 243 ZG. **Article 243V deals with disqualifications: A person shall be disqualified for being chosen as or for being a member of a municipality if he/she is so disqualified:**

(a) under any law for the time being in force for the purposes of elections to the legislature of the state concerned; or (b) under any law made by the state legislature. However; no person shall be disqualified on the ground that he/ she is less than 25 years of age if he/ she has attained the age of 21 years. **If any question arises as to whether a member of a Municipality has become subject to any of the disqualifications mentioned in Clause (1), the question shall be referred for the decision of such authority and in such manner as the Legislature of a State may, by law, provide. Hence, statement 2 is not correct.**

- Q 9. B**
- The SATAT Scheme, which stands for Sustainable Alternative Towards Affordable Transportation, is an initiative led by the Government of India. Its primary objective is to establish Compressed Biogas production facilities and make this eco-friendly fuel available for automotive use. The government aims to achieve this by inviting expressions of interest from potential investors. **Hence statement 1 is correct.**
 - 'SATAT' (Sustainable Alternative Towards Affordable Transportation) scheme on Compressed biogas (CBG) was launched by the Hon'ble Minister, of Petroleum & Natural Gas on 1.10.2018. The scheme envisages to target production of 15 MMT (million tons) of CBG by 2023, from 5000 Plants. **Hence statement 2 is not correct.**
 - To facilitate financing and access to funds to the entrepreneurs, CBG plants have been included under priority sector lending by the Reserve Bank of India. State Bank of India, Punjab National Bank, Canara Bank, Union Bank of India Bank of Baroda etc. have launched products on financing of CBG projects. **Hence statement 3 is correct.**
 - **Hence option (b) is the correct answer.**

- Q 10. A**
- The Sun operates on Nuclear Fusion, not fission. Nuclear fusion is the process where two light atomic nuclei (hydrogen) combine to form a heavier nucleus (helium), releasing energy. Nuclear fission is the splitting of a heavy nucleus into lighter ones, which is the process used in terrestrial nuclear power plants. **Hence, statement 1 is not correct.**
 - Sunspots are cooler than their surroundings, not hotter. Sunspots are areas of intense magnetic activity that inhibit convection, preventing hot gas from the interior from reaching the surface. This makes them roughly 1,500°C to 2,000°C cooler than the rest of the photosphere. Because they are cooler, they emit less light and appear as dark patches. **Hence, statement 2 is not correct.**
 - The Photosphere is the lowest layer of the Sun's atmosphere and is the 'visible surface' we see from Earth. It is the layer where the Sun becomes opaque to visible light, and it is where the majority of the Sun's energy is radiated into space as sunlight. **Hence, statement 3 is correct.**



Q 11. D The students and residents of Calcutta (present-day Kolkata) protested against the 1905 Partition of Bengal in large numbers. In order to suppress the movement forcefully, the then Chief Secretary of Bengal, R W Carlyle, issued a circular in October 1905, which came to be known as the Carlyle Circular. The arbitrary and repressive circular imposed a ban on the participation and collection of people, especially students, in large numbers. Any kind of gatherings, including meetings and demonstrations, were prohibited. Even the chanting of the slogan, Vande Mataram, which had become the rallying cry at the time, was banned.

In response to these arbitrary measures and to protest against the Carlyle Circular, in November 1905, Sachindra Prasad Bose, a student leader of the then Ripon College (present-day Surendra Nath College), founded the Anti-Circular Society. He formed the society with the objective to unite the students and keep the momentum of the movement alive. The society also arranged for alternative education for the students who were expelled from their institutions owing to their participation in the movement. In 1906, Bose also designed a flag. Due to the activities of the society, the nationalistic spirit among the students of Bengal was strong and determined. To restrict the influence of society Bose was arrested in 1906 and sent to the Rawalpindi prison.

Hence, option (d) is the correct answer.

Q 12. C • Researchers from 28 institutions across six States and two UTs have compiled India's first government-supported national biobank for Lysosomal Storage Disorders (LSDs).

• Lysosomes are known as "suicidal bags" because they contain hydrolytic enzymes. If a cell is damaged or aging, these lysosomes can burst, releasing enzymes that digest their own cell, a process called autolysis. **Hence, statement 1 is correct.**

• Lysosomal storage disorders (LSDs) are a group of roughly 50 rare inherited metabolic disorders. They result from genetic defects that lead to a deficiency in specific lysosomal enzymes, causing toxic materials to build up in the body's cells. **Hence, statement 2 is correct.**

• **Currently, there is no known complete cure for most lysosomal storage disorders. While treatments like enzyme replacement therapy (ERT), substrate reduction therapy, and stem cell transplants can manage symptoms and slow disease progression, they do not "cure" the underlying genetic defect. Hence, statement 3 is not correct.**

Q 13. A • Cloud seeding is a weather modification technique that enhances existing precipitation in clouds by dispersing agents like silver iodide or dry ice via aircraft or ground generators. It acts as a catalyst to increase rainfall by from existing, suitable clouds but cannot create rain from a cloudless sky. It works by facilitating moisture accumulation into larger, heavier droplets that fall as rain or snow. Hence, statement 1 is correct.

• Silver iodide has a hexagonal crystalline structure very similar to that of natural ice. This allows it to act as an "ice nucleus," tricking water droplets into freezing around it to form ice crystals, which eventually grow and fall as rain or snow. Hence, statement 2 is correct.

• Weather and clouds occur in the Troposphere (the lowest layer of the atmosphere). The stratosphere is generally very dry and lacks the vertical convection and moisture needed for cloud formation and rain. Cloud seeding is performed on clouds in the troposphere. Hence, statement 3 is not correct.

• "Supercooled" water is liquid water that exists at temperatures below freezing. Cloud seeding (specifically glaciogenic seeding) relies on these droplets. When seeding agents like silver iodide are introduced, these supercooled droplets freeze onto the nuclei, releasing latent heat and triggering the precipitation process. Hence, statement 4 is correct.

- Q 14. B**
- The Netra Mk-II is a next-generation Airborne Early Warning and Control (AEW&C) system developed by India's DRDO for the Indian Air Force, designed to act as a "flying command center" with enhanced surveillance capabilities. Based on modified Airbus A321 aircraft, it features indigenous AESA radar providing 300-360° coverage, significantly improving range (>500 km) and detection over the Netra Mk-I.
 - The program could elevate India into the small circle of nations with homegrown AWACS capabilities, and marks the first time an Airbus platform will take on a role long dominated by Boeing.
 - **Hence, option (b) is the correct answer.**

- Q 15. A**
- **NITI Aayog launched 'Sampoornata Abhiyan', a 3-month campaign from 4th July – 30th September 2024, to achieve saturation in each of the 6 identified indicators in the 112 Aspirational Districts and 500 Aspirational Blocks under the Aspirational Districts Programme and Aspirational Blocks Programme.**
 - **About Aspirational Districts Programme:** Launched in January 2018 by Hon'ble Prime Minister with aims to quickly and effectively transform 112 districts across the country.
 - **Programme Focuses on five themes:**
 - Health & Nutrition
 - Education
 - Agriculture & Water Resources
 - Financial Inclusion & Skill Development
 - Infrastructure
 - NITI Aayog works closely with the respective line Ministries and various development partners to fast-track progress at the district level.
 - **The performance of the Aspirational Districts are available in the public domain portal named "Champions of Change" at championsofchange.gov.in.**
 - **Hence, option (a) is the correct answer.**

- Q 16. C** Process for Admission of New Members to the UN:

Application Submission: The **State submits an application to the Secretary-General, accepting the UN Charter commitments.**

Recommendation from the Security Council requires:

- **9 out of 15 affirmative votes of the Security Council members**
- **With none of the five permanent Council members ("P-5": China, France, Russia, the United Kingdom, the United States) voting against recommendation. Hence, option (c) is the correct answer.**
- General Assembly Approval: Two-thirds majority vote needed if recommended by the Security Council.

- Membership Effectiveness: Membership becomes effective upon adoption of the resolution

Q 17. A Cabinet Committees are not mentioned in the Indian Constitution. They are extra-constitutional bodies, meaning they are not established by the Constitution but exist based on executive orders. **These are constituted under the Transaction of Business Rules, 1961. The cabinet makes use of the committee system to facilitate decision-making in specific areas. Hence, statement 1 is not correct.**

These include cabinet committees on:

- Appointments
- Accommodation
- Economic Affairs
- Parliamentary Affairs
- Political Affairs
- Security
- Investment and Growth
- Skill employment and livelihood

The Political Affairs Committee, Economic Affairs Committee, Appointments Committee are chaired by the Prime Minister and Parliamentary Affairs Committee by the Home Minister. Hence, statement 2 is not correct.

Cabinet Committees can have both Cabinet Ministers and Ministers of State (MoS) as members. **Ministers who are not part of the Union Cabinet may be included in Cabinet Committees depending on their relevance to the committee's function. Hence, statement 3 is correct.**

Q 18. A Central welfare schemes, implemented by the government to provide financial and other aids to individuals who cannot support themselves, can be categorised into two sections — centrally sponsored schemes and central sector schemes.

- Centrally Sponsored Schemes (CSS) are those schemes which are implemented by the state government but sponsored by the central government with a defined shareholding. At present, there are 30 centrally-sponsored schemes but these are umbrella programmes and they have many sub-schemes under them. The funding is borne by the states in the ratio of 50:50 or 90:10 or 75:25 or 70:30. **Hence statement 1 is correct.**
- Centrally Sponsored Schemes are not defined in any article of the Constitution of India. But article 282 enables both the Union and the states to make grants for any public purpose, even if the matter is not within their respective legislative competence. CSSs have been implemented via Article 282 of the Constitution of India. **Hence statement 2 is not correct.**
- CSSs, by their nature, are discretionary transfers made by the Union to the States and routinely pertain to subjects within the State or Concurrent Lists. Effectively, via the instrument of CSSs, the Central Government can determine how and on what activities States are to spend their money. Resultantly, the autonomy which arises from the working of the List system is offset, to an extent, by the manner in which Article 282 is used. **Hence statement 3 is not correct.**

Hence option (a) is the correct answer.

- Q 19. B** • Insurance Regulatory and Development Authority of India (IRDA) regulate the Indian insurance industry to protect the interests of the policyholders and work for the orderly growth of the industry.

Background:

- 1991: Government of India begins the economic reforms programme and financial sector reforms
- 1993: Committee on Reforms in the Insurance Sector, headed by Mr. R. N. Malhotra, (Retired Governor, Reserve Bank of India) set up to recommend reforms.
- 1994: The Malhotra Committee recommends certain reforms having studied the sector and hearing out the stakeholders

Some recommended reforms

- Private sector companies should be allowed to promote insurance companies.
- Foreign promoters should also be allowed.
- Government to vest its regulatory powers on an independent regulatory body answerable to Parliament.

Birth of IRDAI

- Insurance Regulatory and Development Authority (IRDA) set up **as autonomous body under the IRDA Act, 1999. Hence statement 1 is not correct.**

The Authority is a ten member team consisting of:

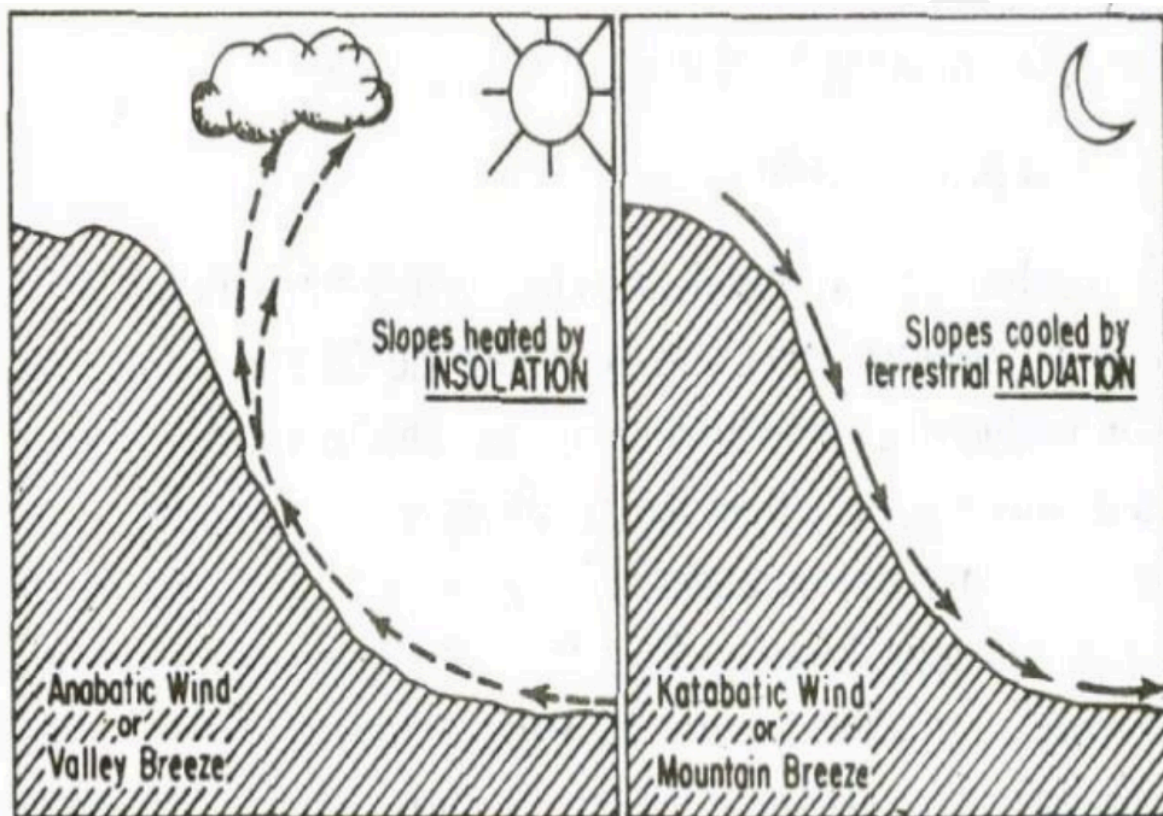
- a Chairman;
- five whole-time members;
- four part-time members,
- **all appointed by the Government of India and Secretary, Department of Financial Services is not its Chairperson. Hence statement 2 is not correct.**

Important Functions:

- To protect the interests of policyholders, to regulate, promote and ensure orderly growth of the insurance industry and for matters connected therewith or incidental thereto.
- Frames regulations for insurance industry in terms of Section 114A of the Insurance Act 1938.
- From the year 2000 has registered new insurance companies in accordance with regulations.
- Monitors insurance sector activities for healthy development of the industry and protection of policyholders' interests.
- **Adjudication of disputes between insurers and intermediaries or insurance intermediaries. Hence statement 3 is correct.**
- To bring about optimum amount of self-regulation in day-to-day working of the industry consistent with the requirements of prudential regulation.

- Q 20. A** • The evolved Laser Interferometer Space Antenna (eLISA) is a space-based observatory designed by the European Space Agency (ESA) to measure gravitational waves, tiny ripples in spacetime caused by massive cosmic events. Unlike ground-based detectors like LIGO, eLISA can detect low-frequency waves that are blocked by Earth's seismic noise, allowing it to study phenomena such as merging supermassive black holes at the centers of galaxies and thousands of compact binary star systems. It uses three spacecraft arranged in a triangular formation, separated by millions of kilometers, using laser interferometry to track minuscule changes in their relative distance.
- **Hence, option (a) is the correct answer.**

- Q 21. A**
- Anabatic winds (from the Greek word anabatikos, meaning "going upward") are localized mountain winds that occur specifically during the daytime. **Hence, statement I is correct.**
 - During the day, the sun heats the mountain slopes more rapidly than the air at the same altitude in the valley or over the plains. The air in direct contact with these slopes becomes warm, less dense, and buoyant. **Hence, statement II is correct.**
 - Because this air is warmer and lighter than the air further away from the slope, it begins to rise upslope. This upward movement of air is the definition of an Anabatic wind. Therefore, the solar heating described in **Statement-II** is the direct cause and explanation for the phenomenon in Statement-I.
 - Hence, option (a) is the correct answer.**



- Q 22. A**
- High government borrowing can lead to an increase in interest rates if the government competes with the private sector for limited funds in the financial markets. This makes it more expensive for private businesses to borrow, potentially reducing private investment. **Hence option 1 is correct.**
 - If high government borrowing is financed by printing more money (monetary financing), it can lead to higher inflation as it increases the money supply.**
 - Further high government borrowing results in increased spending on welfare schemes and public investment in infrastructure. This leads to increased money supply in the economy and high inflation. **Hence option 2 is not correct.**
 - Increased government borrowing can lead to increased supply of domestic currency in the forex market.**
 - If government borrowing increases, the fiscal deficit directly increases because the fiscal deficit represents total borrowing requirements. Consequently, the primary deficit also increases, as it is calculated as the fiscal deficit minus interest payments, reflecting a higher need for funds to cover current expenses. **Hence option 3 is not correct.**

Hence option (a) is the correct answer.

- Q 23. A** • **MSME stands for Micro, Small and Medium Enterprises.** MSMEs are small businesses with limited operations and budgets that play a vital role in the Indian economy. They Contribute around 30% to India's GDP, ~46% of India's total exports.
- **MSMEs are broadly classified into Micro, Small and Medium based on their investment in plant and machinery for manufacturing or equipment for service enterprises, as well as their annual turnover. Hence, statement 2 is correct.**
 - **The Micro, Small and Medium Enterprises Development (MSMED) Act, 2006 regulates micro, small, and medium enterprises (MSMEs) in India. The MSMED Act was created to promote, develop, and enhance the competitiveness of MSMEs. It is the principal legislation that governs the registration, definition, and development of MSMEs Hence, statement 1 is not correct.**
 - **The Ministry of Micro, Small and Medium Enterprises is the apex executive body for the formulation and administration of rules, regulations, and laws relating to micro, small and medium enterprises in India. Hence, statement 3 is not correct.**
 - **Hence, option(a) is the correct answer.**

- Q 24. A** • Directed Energy Weapons (DEWs) are advanced military systems that utilize concentrated electromagnetic energy or particle beams to incapacitate, damage, or destroy enemy equipment, facilities, or personnel. Unlike traditional weapons that rely on physical projectiles or explosive force, DEWs deliver energy at the speed of light, allowing for near-instantaneous engagement of targets.

Common types include:

- **High-Energy Lasers (HEL):** Use concentrated light to burn or melt targets like drones and missiles.
 - **High-Power Microwaves (HPM):** Emit radio-frequency pulses to disrupt or destroy electronic circuits and sensors.
 - **Particle Beam Weapons:** Accelerate subatomic particles to near-light speeds to cause damage **at the atomic level.**
- **Hence, option (a) is the correct answer.**

- Q 25. A** • **Tundra Biome:** This biome has very low primary productivity due to extreme cold, permafrost, and a very short growing season. It contributes the least amount of oxygen globally.
- **Temperate Deciduous Forest:** These forests have moderate productivity. While they are significant, they cover less land area and have lower photosynthetic rates annually than tropical forests.
 - **Tropical Rainforest:** Often called the "lungs of the Earth," these are the most productive terrestrial biomes. They contribute a massive amount of oxygen (~20-30% of terrestrial production), but their net contribution is often debated because they also consume a lot of oxygen through the decay of organic matter.
 - **Open Ocean Phytoplankton:** Marine organisms (phytoplankton, algae, and cyanobacteria) are the largest contributors to the world's oxygen. It is estimated that they produce 50% to 80% of the Earth's oxygen because they cover 70% of the planet's surface and have high turnover rates.

Hence, option (a) is the correct answer.

- Q 26. B** • A quasi-moon is not a natural satellite permanently bound by Earth's gravity. Unlike our actual Moon, which is gravitationally bound to Earth, quasi-moons are temporary and orbit the Sun. They only appear to orbit Earth from our perspective. **Hence, statement 1 is not correct.**
- PN7 revolves around the Sun in a 1:1 orbital resonance with Earth. This co-orbital motion makes it look like it is orbiting Earth when viewed from a terrestrial frame of reference. **Hence, statement 2 is correct.**
 - Quasi-moons like 2025 PN7 are part of the Near-Earth Object (NEO) population. Specifically, PN7 belongs to the "Arjuna" class of asteroids, which have orbits very similar to Earth's. **Hence, statement 3 is correct.**
 - By definition, a quasi-moon lies outside Earth's Hill sphere (the region where a planet's gravity dominates over the Sun's). Because it is outside this sphere, it is not gravitationally bound to Earth, which is why it eventually drifts away after several decades or centuries. **Hence, statement 4 is not correct.**
- Q 27. A** • As per article 167 of the Constitution of India, It shall be the duty of the Chief Minister of each State—
- (a) to communicate to the Governor of the State all decisions of the Council of Ministers relating to the administration of the affairs of the State and proposals for legislation;
 - (b) to furnish such information relating to the administration of the affairs of the State and proposals for legislation as the Governor may call for. **Hence, statement 2 is correct.**
 - (c) if the Governor so requires, to submit for the consideration of the Council of Ministers any matter on which a decision has been taken by a Minister but which has not been considered by the Council. **Hence, statement 3 is correct.**
- As per article 167 (1) of the Constitution of India, the Chief Minister shall be appointed by the Governor and the other Ministers shall be appointed by the Governor on the advice of the Chief Minister, and the Ministers shall hold office during the pleasure of the Governor. **Hence, statement 3 is not correct.**
- Q 28. A** Agni-III is an Intermediate-Range Ballistic Missile (IRBM) that was designed and developed by the Defence Research and Development Organisation (DRDO). The Agni-III is a ballistic missile (IRBM) with a range of 3,000 – 3,500 km. It entered service with India's Strategic Forces Command in 2011, serving alongside the Agni-II as a nuclear delivery system. **Hence, statement 1 is correct but statement 3 is not correct.**
- Agni-III is a two-stage missile that uses solid propellant in both of its propulsion stages, not liquid propellant. Solid fuel allows for easier storage and faster launch times compared to liquid-fueled missiles like the earlier Prithvi series. **Hence, statement 2 is not correct.**
- Q 29. A** • Sir Andrew Fraser is related to police reforms. He chaired the Police Commission of 1902–03 during Lord Curzon's viceroyalty. The commission's recommendations led to significant changes,

including the establishment of the Criminal Investigation Department (CID) at both the central and provincial levels. **Hence, point 1 is correct.**

- Butler Commission (1927): Also known as the Indian States Committee, it was formed to examine the relationship between the British Paramount Power and the Indian Princely States. **Hence, point 2 is not correct.**
- Hartog Committee (1929): This committee focused on education in British India, highlighting the issues of "wastage" and "stagnation" in the primary education system. **Hence, point 3 is not correct.**
- Lothian Committee (1932): Officially the Indian Franchise Committee, it was established to determine the electoral franchise (voting rights) for the new constitutional setup following the Round Table Conferences.

- Q 30. B**
- As magma rises at the mid-ocean ridge and cools, it records the Earth's magnetic polarity. Scientists discovered that the magnetic "stripes" on one side of the ridge are a mirror image of those on the other side. This proves that the ocean floor is moving outward from the center like a conveyor belt. **Hence, statement 1 is correct.**
 - Radiometric dating shows that the rocks located exactly at the crest of the mid-ocean ridge are the youngest. As you move farther away from the ridge toward the continents, the rocks become progressively older. This confirms that new crust is being created at the ridge and pushed aside. **Hence, statement 2 is correct.**
 - **Presence of identical fossils on the coastlines of Africa and South America:** While this is a major piece of evidence, it supports Alfred Wegener's Continental Drift Theory, not the Sea Floor Spreading Theory. Continental Drift was based on matching coastlines and fossil distribution (like Mesosaurus), whereas Sea Floor Spreading focused on the processes occurring at the ocean floor itself. **Hence, statement 3 is not correct.**

Q 31. D 0° (Prime Meridian) – Spain

- The Prime Meridian (0° longitude) is the reference line for Greenwich Mean Time (GMT) and passes through several countries, including:
- United Kingdom (Greenwich, London), France, **Spain**, Algeria, Mali, Burkina Faso, Ghana. **Hence pair 1 is correctly matched.**



180° Meridian – Fiji Islands

- The 180° Meridian is located directly opposite the Prime Meridian, and while it generally follows the International Date Line (IDL), it deviates to accommodate international boundaries. Countries and regions it passes through include:
- **Fiji Islands** (deviation to keep the islands in the same time zone. The IDL is on Taveuni in Fiji).
- Russia (easternmost part),
- Kiribati, Tuvalu. **Hence pair 2 is correctly matched.**

Tropic of Cancer – Bangladesh

- The Tropic of Cancer is a line of latitude approximately 23.5° north of the equator. It passes through the following countries
- **Americas:** Mexico, Bahamas,
- **Africa:** Western Sahara, Mauritania, Mali, Algeria, Niger, Libya, Egypt,
- **Asia:** Saudi Arabia, United Arab Emirates, Oman, India, **Bangladesh**, Myanmar, China, Taiwan. **Hence pair 3 is correctly matched.**

Tropic of Capricorn – Botswana

- The Tropic of Capricorn is located at 23.5° south of the equator. It crosses the following countries:
- Chile, Argentina, Paraguay, Brazil, Namibia, **Botswana**, South Africa, Madagascar. **Hence pair 4 is correctly matched.**

- Q 32. C • In accordance with the goal of the state controlling the commanding heights of the economy, the Industrial Policy Resolution of 1956 was adopted. **This resolution formed the basis of the Second Five Year Plan**, the plan which tried to build the basis for a socialist pattern of society. **Hence, statement 1 is not correct.**

This resolution classified industries into three categories.

- the first category comprised industries that would be exclusively owned by the state
- **the second category consisted of industries in which the private sector** could supplement the efforts of the state sector, with the state taking the sole responsibility for starting new units
- **the third category consisted of the remaining industries which were to be in the private sector; Hence, statement 2 is not correct.**
- Although there was a category of industries left to the private sector, the sector was kept under state control through a system of licenses. No new industry was allowed unless a license was obtained from the government.
- **This policy was used for promoting industry in backward regions;** it was easier to obtain a license if the industrial unit was established in an economically backward area. In addition, such units were given certain concessions such as tax benefits and electricity at a lower tariff. The purpose of this policy was to promote regional equality. **Hence, statement 3 is correct.**

- Q 33. C • The Faraizi Movement (1818) was a 19th-century Islamic reformist and agrarian movement in Bengal, founded by Haji Shariatullah to purify Islam by removing un-Islamic practices (shirk/bida) and enforcing obligatory duties (faraiz). Under Dudu Miyan, it became a significant, militant anti-zamindar and anti-British peasant revolt, fighting against illegal taxes. **Hence, statement 1 is correct.**

- The Faraizi movement was a revivalist and traditionalist movement rather than a modernist one. It did not aim to introduce Western education; instead, it focused on returning to the basic tenets of the Quran and Hadith. Modernisation through Western education was a hallmark of other movements, such as the Aligarh Movement led by Sir Syed Ahmed Khan. **Hence, statement 2 is not correct.**
- While it began as a religious reform, it evolved into a significant agrarian movement. Under the leadership of Haji Shariatullah's son, Dudu Miyan, the movement organised the Muslim peasantry against the exploitation of both Hindu and British landlords and indigo planters. Dudu Miyan famously preached that land belongs to God and no one has the right to charge rent for it. **Hence, statement 3 is correct.**

- Q 34. A**
- Tropical rainforests are incredibly complex ecosystems that rely on a delicate cycle of nutrients. When large-scale deforestation occurs (especially when land is cleared for farming or cattle ranching), the cycle is broken. Unlike temperate forests, rainforests often struggle to "naturally" regrow their original structure and biodiversity because the conditions that sustained them **have been stripped away. Hence, Assertion (A) is correct.**
 - In a tropical rainforest, the soil is surprisingly nutrient-poor. This is due to heavy rainfall which "leaches" (washes away) minerals and nutrients from the soil. Instead, the nutrients are rapidly taken up by the dense vegetation. Therefore, almost all the "capital" or nutrients of the ecosystem are stored in the biomass (the trees and plants themselves). **Hence, Reason (R) is correct.**
 - Since the nutrients are stored in the trees, removing the trees means removing the entire nutrient supply for that area. Without the trees to provide leaf litter (which decays quickly and provides "instant" food for other plants), the soil becomes infertile almost immediately. Because the soil cannot support new growth, the forest fails to regenerate naturally after large-scale clearing.
 - **Hence, option (a) is the correct answer.**

- Q 35. C**
- The advent of the Hydrocarbon Exploration and Licensing Policy (HELP) promulgated in March 2016, brought about paradigm changes in the way exploratory blocks are auctioned in India.
 - **The Open Acreage Licensing Policy (OALP), launched in 2016 under the Hydrocarbon Exploration and Licensing Policy (HELP), offers the potential investors freedom to carve out blocks of their choice through submission of Expression of Interest (EoI). The EoI window is open round the year and the investors need to wait for periodic launch of bid rounds. Hence, statement 1 is correct.**
 - The OALP, a critical part of the Hydrocarbon Exploration and Licensing Policy, **provides uniform licences for exploration and production of all forms of hydrocarbons, enabling contractors to explore conventional as well as unconventional oil and gas resources. Fields are offered under a revenue-sharing model and throw up marketing and pricing freedom for crude oil and natural gas produced. Hence, statement 2 is not correct, but 3 is correct.**

Hydrocarbon Exploration and Licensing Policy (HELP)

- The Hydrocarbon Exploration & Licensing Policy (HELP), which adopts the Revenue Sharing Contract model, is a giant step towards improving the 'Ease of Doing Business' in the Indian Exploration and Production (E&P) sector.
- It comes with attractive and liberal terms like reduced royalty rates, no Oil Cess, marketing and pricing freedom, round the year bidding, freedom to investors for carving out blocks of their interest, a single license to cover both conventional and unconventional hydrocarbon resources,

exploration permission during the entire contract period, and an easy, transparent and swift bidding and awarding process.

- Q 36. B** • Wearable devices rely on embedded sensors to monitor a wide range of user data, including physiological parameters (heart rate, blood oxygen levels, body temperature) and physical parameters (steps, motion, posture). Hence, statement 1 is correct.
- These technologies are designed with wireless connectivity (such as Bluetooth, Wi-Fi, or cellular networks) to transmit collected data to smartphones, tablets, or cloud servers for further analysis and long-term storage. **Hence, statement 2 is correct.**
 - Modern wearable ecosystems use AI and machine learning algorithms to analyze patterns in collected data, which can predict potential health issues like irregular heart rhythms (AFib) or onset of illness. **Hence, statement 3 is correct.**
 - While many sensors (like heart rate monitors) require skin contact, the word "must" makes this statement false. Some wearable technologies, such as smart glasses or certain motion-sensing clothing, do not require continuous contact with the skin to perform their functions (e.g., GPS tracking, visual overlays, or environment sensing). **Hence, statement 4 is not correct.**
- Q 37. C** • Tax buoyancy accounts for the total change in tax revenue as a result of both automatic responses to changes in the economic base and any discretionary tax policy changes e.g. rate increases or the introduction of new taxes. This makes tax buoyancy a broader measure than tax elasticity
- Tax elasticity measures the responsiveness of tax revenue to changes in the economic base, such as GDP growth, without any changes in tax rate or policy. It isolates the effect of economic growth on revenue, assuming tax policies remain constant
 - While a higher tax elasticity can contribute to higher tax buoyancy, it does not guarantee it. If tax policies or rates remain unchanged, a country could have high tax elasticity without seeing an increase in tax buoyancy. Alternatively, a country with low tax elasticity could achieve high tax buoyancy through frequent or effective policy changes. **Hence, option (c) is the correct answer.**
- Q 38. C** • **Financial centres that cater to customers outside their own jurisdiction** are called **International Financial Service Centre (IFSC)** or **offshore Financial Centers (OFCs)**. All these centres are considered 'international' and they deal with the flow of finance and financial products/services across borders. **Hence, statement 1 is correct.**
- The **Gujarat International Finance Tec-City (GIFT City)** multi services special economic zone (SEZ) has set up the **first International Financial Service Centre in India (IFSC)** in accordance with the **SEZ Act 2005**. **Hence, statement 2 and 3 are correct.**
 - The **IFSC in GIFT City (IFSC-GIFT)** is being developed as a global financial and information technology services hub designed to be at or above par with globally benchmarked financial centres such as London, Hong Kong, Singapore, and Dubai. Thus, it seeks to bring to India, those types of financial services and transactions that are currently carried on outside India by overseas financial institutions and overseas branches or subsidiaries of Indian financial institutions.
- Q 39. C** • Exercise MITRA SHAKTI is a bilateral joint military exercise between the Indian Army and the Sri Lankan Army. It has been conducted annually (alternating between both countries) since its

inception in 2012. It is designed to foster closer relations between the two armies and enhance their ability to conduct joint operations. **Hence, statement 1 is correct.**

The 11th edition of this exercise (MITRA SHAKTI-2025) was hosted by India.

- The location for this specific drill was the Maratha Light Infantry Regimental Centre in Belagavi, Karnataka.
- This edition was significant for its focus on modern warfare technologies, specifically the integration of Drones and Counter-Unmanned Aerial Systems (C-UAS).

• **Hence, statement 2 is correct.**

Q 40. B • Near the Coast, both speed and wavelength decrease, not increase. This phenomenon is known as Wave Shoaling. As a tsunami enters shallower water near the coast, friction with the rising seabed causes the wave to slow down (speed decreases). As the front of the wave slows, the faster-moving water behind it bunches up, causing the wavelength to decrease and the wave height (amplitude) to increase dramatically. **Hence, statement 1 is not correct.**

- In the deep, open ocean, a tsunami travels at extremely high speeds (often over 700 km/h) but has a very small amplitude, usually less than 1 meter. However, its wavelength is incredibly long, sometimes stretching over 200 kilometers. This makes it almost unnoticeable to ships at sea. **Hence, statement 2 is correct.**

- **When a tsunami travels, it acts as a shallow-water wave. While the Coriolis effect does play a tiny role in the propagation of very long-distance tsunamis (teletsunamis), it does not impact the initial strength, height, or speed of the waves at the source. Hence, statement 3 is not correct.**

Q 41. A • **Net National Income is Net National Product at factor cost.** It represents the total income earned by a country's residents (wages, rent, interest, profit) from producing goods and services, calculated after adjusting for depreciation and removing net indirect taxes. It is considered the official measure of national income.

- It serves as the official measure of a country's national income, representing the total earnings of residents (including wages, rent, interest, and profits) from the production of goods and services.

To find Net National Income at Factor Cost (NNP_{at FC}), we follow the following steps:

Convert GDP at Market Price to NDP at Market Price

$$\text{NDP(MP)} = \text{GDP(MP)} - \text{Depreciation} = 80,000 - 5,000 = 75,000$$

Now, to move from Domestic to National, add Net Factor Income from Abroad (NFIA)

$$\text{NNPMP} = \text{NDPMP} + \text{NFIA} = 75,000 + (-2,000) = 73,000$$

Next, the market price includes indirect taxes and excludes subsidies. To get Factor Cost, we subtract Net Indirect Taxes.

$$\text{NNP at Factor Cost} = \text{NNP at Market Price} - \text{Net Indirect Taxes} = 73,000 - 3,000 = 70,000.$$

Hence, option (a) is the correct answer.

Q 42. A **Sukinda Valley**

- State: Odisha (Jajpur district).
- Significance: This valley is world-famous for harboring approximately 95% of India's total chromite (chromium ore) resources. It is one of the largest open-cast chromite mining areas in the

world. Hence, row 1 is correctly matched.

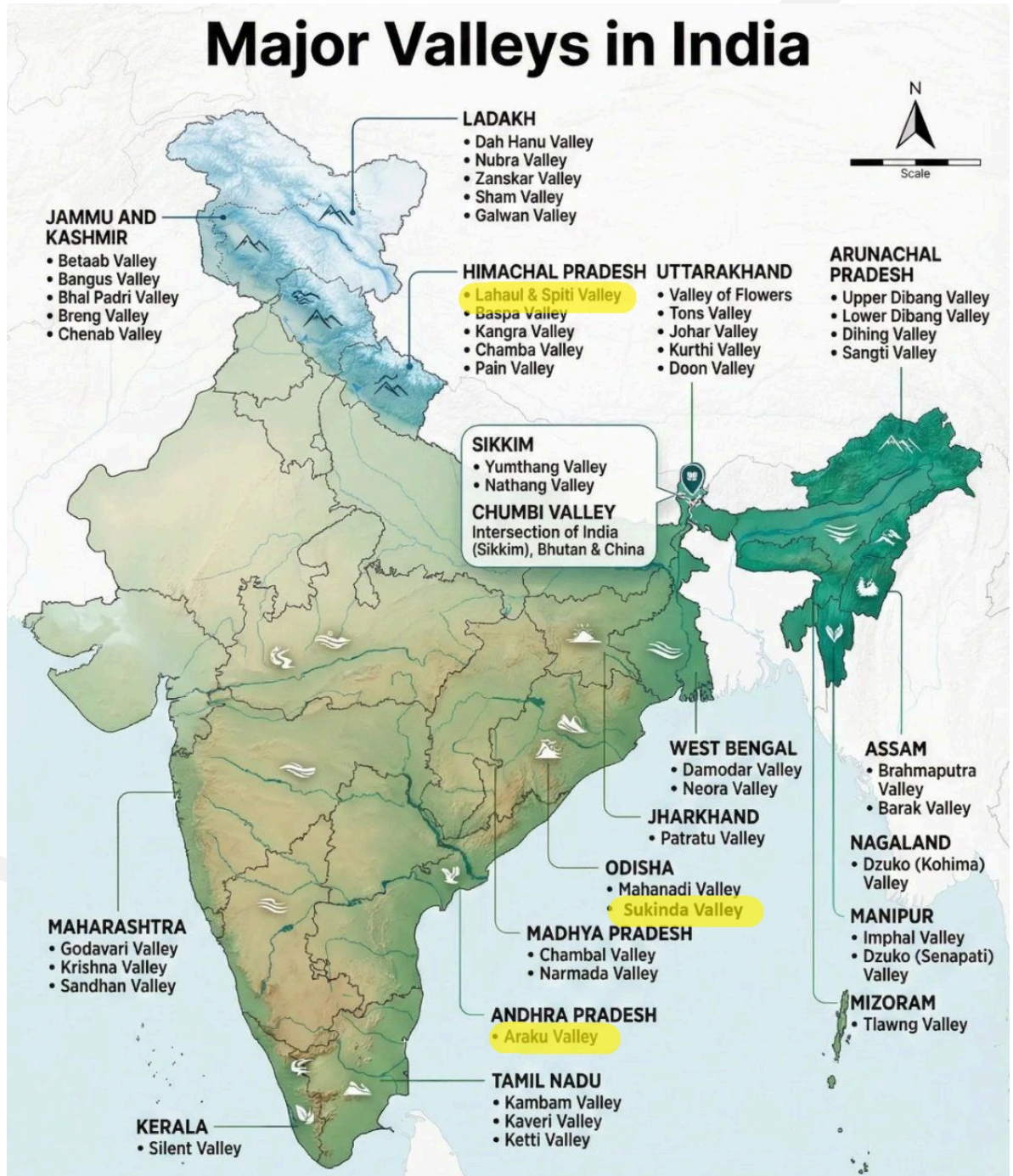
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Lahaul Valley

- o **Lahaul Valley is located in Himachal Pradesh.** While it is a high-altitude cold desert region, it is administratively and geologically distinct from the Kashmir Valley. It is connected to the Spiti Valley via the Kunzum Pass. **Hence, row 2 is not correctly matched.**

Araku Valley

- o **Araku Valley is a famous hill station and valley located in the Eastern Ghats of Andhra Pradesh** (Alluri Sitharama Raju district). While it is correctly known for its coffee cultivation (organic Araku Coffee), the state matching is wrong. **Hence, row 3 is not correctly matched.**



- Q 43. C • Solar - Rajasthan: Rajasthan holds the highest installed solar power generation capacity in India, exceeding 33 GW. Due to its vast desert land and high solar irradiance, Rajasthan leads in large-

scale solar projects, followed closely by Gujarat and Karnataka. It hosts the Bhadla Solar Park, one of the world's largest.

- **Wind - Gujarat:** Gujarat has the highest installed wind power capacity in India, having overtaken Tamil Nadu with over 13,800 MW to 14,000+ MW of installed capacity. The state is a leader in onshore wind energy, aided by massive projects in the Kutch region.
- **Biomass - Maharashtra:** Maharashtra consistently leads in biomass and bagasse cogeneration capacity due to its extensive agricultural base and large number of sugar mills.
- **Small Hydro - Karnataka:** Karnataka holds the highest installed capacity for Small Hydro Power (projects up to 25 MW) in India, supported by its hilly terrain and river systems.

Hence, option (c) is the correct answer.

Q 44. C • Recently, the State-owned Bank of India on Tuesday said it has raised funds through issue of long-term infrastructure bonds of Rs 10,000 crore at a coupon rate of 7.23 per cent per annum. With strong demand and regulatory advantages, total infrastructure bond issuances were expected to surpass Rs 1.50 lakh crore in FY26.

- Infrastructure bonds are long-term, fixed-income debt securities issued by government-backed entities, PSUs, or private corporations to fund major projects like roads, railways, and power plants. They provide investors with steady, predictable income and potential tax benefits, often acting as a safe, low-risk investment for conservative portfolios, with maturities usually ranging from 10 to 15 years.
- The proceeds from these bonds are specifically earmarked for infrastructure development. By issuing these bonds, entities can raise the necessary capital to invest in essential public assets without immediate financial strain. Investors, in turn, receive a stable and lucrative return on their investment.

Benefits of Investing in Infrastructure Bonds

- **Stable Returns:** Infrastructure bonds typically offer fixed interest rates, providing a predictable income stream for investors.
- **Long-Term Investment:** With maturities often ranging from 10 to 15 years, these bonds are ideal for long-term investors looking to diversify their portfolios.
- **Portfolio Diversification:** By including these bonds, investors can balance their portfolio risk, particularly during volatile market conditions. This diversification helps in spreading risk across different asset classes and reduces the impact of market fluctuations.
- **Contribution to National Growth:** By investing in these bonds, individuals can contribute to the development of critical infrastructure, aiding in national growth and development.

Risks of investing in Infrastructure Bonds

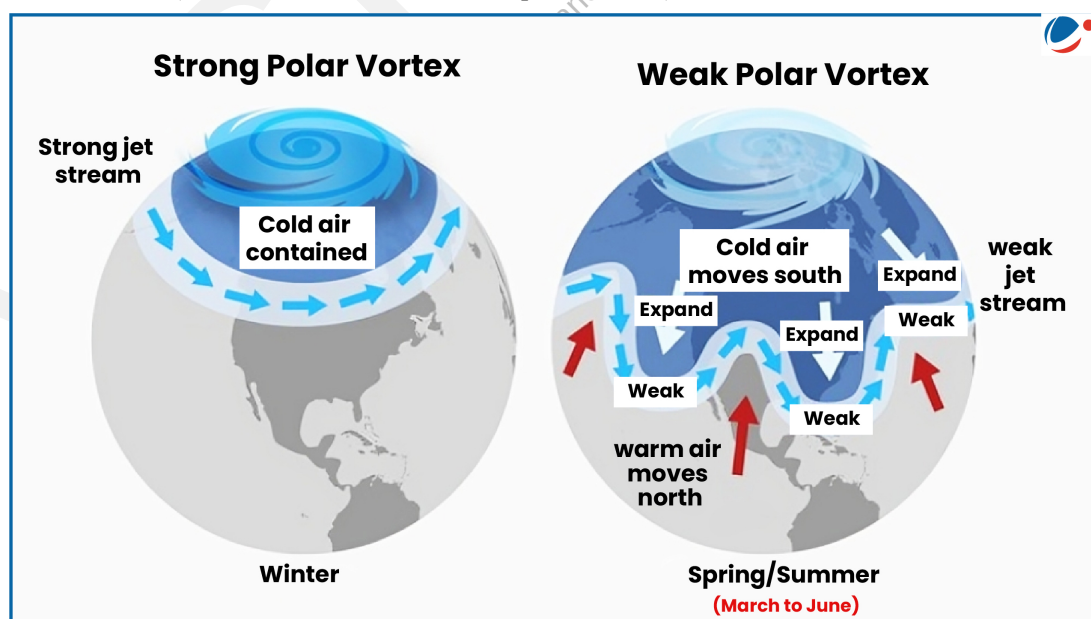
- **Interest Rate Risk:** As with any fixed-income security, the value of infrastructure bonds can be affected by changes in interest rates. Hence, statement II is not correct.
- **Liquidity Risk:** These bonds may not be as liquid as other investments, making it difficult to sell them before maturity.
- **The government has encouraged banks to raise infrastructure-dedicated bonds, which fund infrastructure projects, affordable housing, renewables, and more. These bonds offer banks a cheaper way of raising money since they don't have to maintain Cash Reserve Ratio (CRR) or**

Statutory liquidity Ratio (SLR) required for traditional deposits. Hence, Banks have significant headroom to expand infrastructure bond issuances. Hence, statement III is correct and it explains the statement I.

- Also, infrastructure bonds enjoy senior unsecured status, making them a safer option than subordinated tier-2 and additional tier-1 (AT-1) bonds. **Hence, option (c) is the correct answer.**

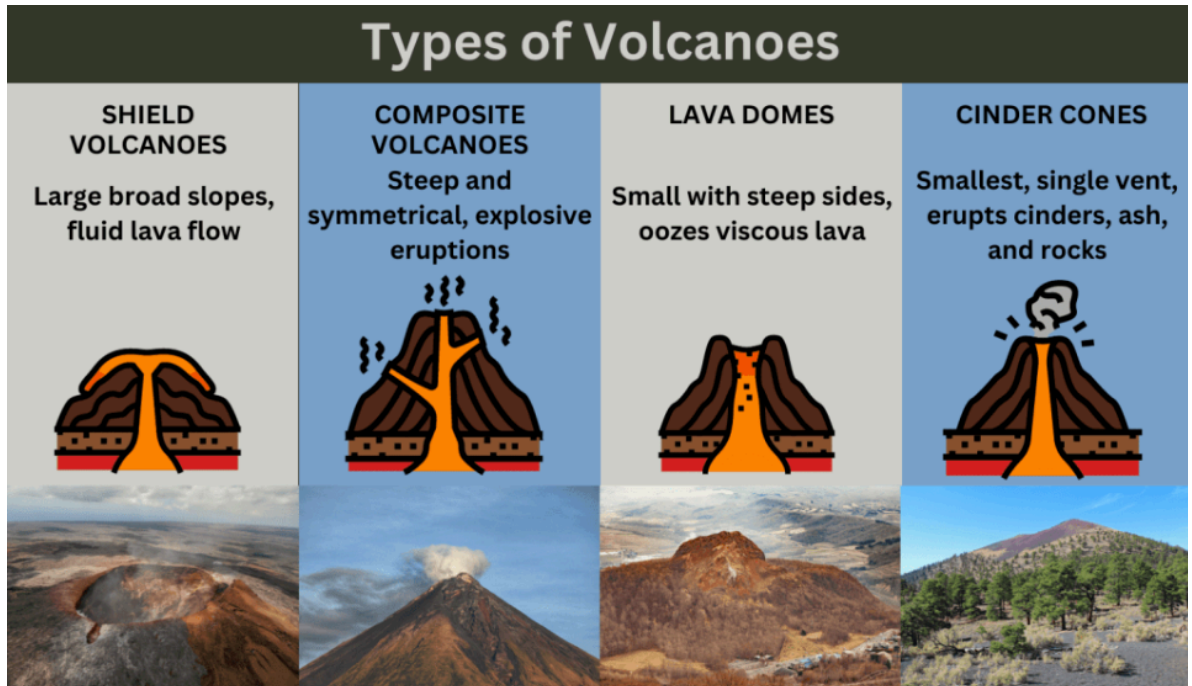
- Q 45. B**
- Announced in the Union Budget 2026-27, Bharat-VISTAAR (Virtually Integrated System to Access Agricultural Resources) is a cutting-edge digital public infrastructure designed to revolutionize the agricultural sector.
 - Core Technology:** It is a multilingual AI tool that integrates data from AgriStack (digital farmer records) and the Indian Council of Agricultural Research (ICAR). **Hence, option (b) is the correct answer.**
 - Functionality:** It serves as a 24/7 "digital agriculture expert," accessible via a dedicated helpline (155261) or mobile app. Its conversational AI assistant, Bharati, provides real-time, hyper-local advice in over 22 regional languages.
 - Key Services:** The platform offers customized information on crop planning, pest management, weather updates, market trends, and access to 10 major government schemes like PM-KISAN and Kisan Credit Card.

- Q 46. C**
- In the Northern Hemisphere, the Polar Vortex is a large area of low pressure and cold air that rotates counterclockwise due to the Coriolis effect. It typically stays centered over the North Pole, held in place by the polar jet stream. **Hence, statement 1 is correct.**
 - An Sudden Stratospheric Warming (SSW) event involves a rapid increase in temperatures in the stratosphere, which disrupts the stable, westerly winds of the polar vortex. This disruption causes the vortex to weaken, break, or become displaced (shift) southward, leading to cold outbreaks in the mid-latitudes (such as North America, Europe, and Asia). **Hence, statement 2 is correct.**



- Q 47. B**
- Shield volcanoes are composed almost entirely of fluid, basaltic lava flows. Basaltic lava has low viscosity (it is very runny). This allows the lava to travel long distances from the vent, creating a broad, gently sloping cone that resembles a warrior's shield. **Hence, statement 1 is correct.**

- Composite volcanoes (or stratovolcanoes) are typically much more explosive than shield volcanoes. They are built from 'andesitic' or 'rhyolitic' lavas, which have high viscosity. This thick lava traps volcanic gases, leading to a massive buildup of pressure that eventually results in violent, explosive eruptions. **Hence, statement 2 is correct.**
- Hotspots do not occur only at convergent boundaries. Hotspots are intra-plate phenomena. They occur where a plume of hot mantle material rises through the middle of a tectonic plate, far away from any boundary.
- Examples: The Hawaiian Islands are the classic example of hotspot volcanism occurring in the middle of the Pacific Plate, not at a convergent boundary. **Hence, statement 3 is not correct.**



- Q 48. C**
- Akbar introduced the Mansabdari system in his administration. Mansabdari system was an administrative system which look after the civil and military affairs of the state. **Hence, statement 1 is correct.**
 - Under this system every officer was assigned a rank (mansab). The lowest rank was 10 and the highest was 5000 for the nobles. Princes of royal blood received even higher ranks. The ranks were divided into two – zat and sawar. Zat means personal and it fixed the personal status of a person. Sawar rank indicated the number of cavalymen of a person who was required to maintain. Every sawar had to maintain at least two horses. **Hence, statement 2 is correct.**
 - The mansab rank was not hereditary. All appointments and promotions as well as dismissals were directly made by the emperor. **Hence, statement 3 is not correct.**

- Q 49. A** Article 39A of the Indian Constitution (added by the 42nd Amendment, 1976) explicitly provides for free legal aid to ensure justice for all, regardless of economic or other disabilities. **Hence, statement 1 is correct.**

Additionally, the right to free legal aid has been upheld by the Supreme Court as part of Article 21 (Right to Life and Personal Liberty) in cases such as Hussainara Khatoon v. State of Bihar (1979) and Khatri v. State of Bihar (1981).

Legal aid is not limited to individuals. It can also be provided to groups, organizations, and public interest litigations (PILs) when their cause aligns with justice and social welfare. For example, legal

aid can be extended to NGOs, labor unions, and marginalized community groups seeking justice. **Hence, statement 2 is not correct.**

Free legal aid is the provision of free legal services in civil and criminal matters for those poor and marginalized people who cannot afford the services of a lawyer for the conduct of a case or a legal proceeding in any Court, Tribunal or Authority. These services are governed by Legal Services Authorities Act, 1987 and headed by the National Legal Services Authority (NALSA). **Hence, statement 3 is not correct.**

Q 50. B Kuchipudi is one of the Indian classical dances belonging to Andhra Pradesh. It was originally performed by a group of itinerant actors known as Bhagavathalu, who would travel from village to village, performing plays and dances based on Hindu mythology. **Hence, statement 1 is not correct.**

In the 15th century, the great poet and musician, Siddhendra Yogi, played a major role in the development of Kuchipudi. He is credited with transforming the dance form from a simple folk art to a sophisticated and refined classical dance form.

Features

- The dance form is characterized by its intricate footwork, graceful movements, and subtle facial expressions.
- **Kuchipudi originated as a dance-drama (Yakshagana) performed by groups of male actors (Bhagavatulu). The solo dance tradition is a modern evolution of the art form. Hence, statement 2 is not correct.**
- **It incorporates both pure dance (Nritya) and expressive dance (Nritya) elements, as well as storytelling through dance (Natya). Hence, statement 3 is correct.**
- It is also performed on the edge of a brass plate (known as Tarangam) on the beats of Carnatic music.
- **It is largely developed as a Hindu god Krishna-oriented Vaishnavism tradition, and it is most closely related to Bhagavata Mela. Hence, statement 4 is correct.**
- The Kuchipudi performer apart from being a dancer and actor has to be skilled in Sanskrit and Telugu languages, music and manuscripts of the performance.
- The Kuchipudi dancers wear light make-up and ornaments like the Rakudi (head ornament), Chandra Vanki (arm band), Adda Bhasa and Kasina Sara (necklace). A long plait is decorated with flowers and jewelry. The ornaments are made of light wood called Boorugu.
- The typical musical instruments in Kuchipudi are mridangam, cymbals, veena, flute and the tambura.

Q 51. A A specific type of butterfly commonly found in South Asia and Southeast Asia:

- **Common Crow:** Scientifically known as *Euploea core*, it is a glossy black butterfly with white spots. It is known for being unpalatable to predators.
- **Plain Orange Tip:** Scientifically known as *Colotis aurora*, this butterfly is part of the Pieridae family and is easily recognised by the bright orange patches on its wingtips.
- **Blue Tiger:** Scientifically known as *Tirumala limniace*, this is a large butterfly with dark wings featuring pale blue semi-transparent streaks and spots. They are well-known for their migratory behaviour.

- These species are often cited in news reports related to biodiversity counts, seasonal migrations, or environmental conservation efforts.

Hence, option (a) is the correct answer.

- Q 52. B**
- According to WTO, non-market economy refers to a particular situation of economies where the government has a complete or substantially complete monopoly of its trade and where all domestic prices are fixed by the State.
 - General Agreement on Tariffs and Trade (GATT) 1994 recognizes that a strict comparison with home market prices may not be appropriate and thus importing countries can exercise significant discretion in the calculation of the normal value of products exported from non-market economies. **Therefore, World Trade Organization (WTO) members are allowed to apply anti-dumping measures on any company if it exports a product at a lower price than its home market, and if the product threatens to impact the local industry. Hence, statement 2 is correct.**
 - China joined the WTO in December 2001. The European Union and the US have desisted from granting market economy status to China, citing wide-ranging price control on export commodities by China. India also rejected China's demand to grant it market economy status and will continue to treat China as a non-market economy. **Market economy status is granted to the country by the WTO member countries. Hence, statement 1 is not correct.**

- Q 53. A**
- Mahabhashya (meaning "Great Commentary") is a monumental Sanskrit grammatical text authored by Patanjali. It is primarily an extensive commentary on Panini's Ashtadhyayi, the foundational treatise of Sanskrit grammar. Patanjali also analyses and sometimes critiques the Varttikas (critical notes) written by the grammarian Katyayana on Panini's rules. The "Munitraya": Together with Panini and Katyayana, Patanjali is revered as one of the three sages (Munitraya) who established the authoritative structure of the Sanskrit language. **Hence, Pair 1 is not correctly matched.**
 - Mālavikāgnimitram: It (Malavika and Agnimitra) is the first drama written by the celebrated Sanskrit poet Kālidāsa, likely in the 5th century CE. Unlike his more spiritual or epic works, this five-act play is a lighthearted romantic comedy set within a royal harem. Mālavikāgnimitram tells the story of the love of Agnimitra, the Shunga Emperor at Vidisha, for the beautiful handmaiden of his chief queen. **Hence, Pair 2 is correctly matched.**
 - The Aṣṭādhyāyī is a foundational Sanskrit grammar treatise written by the ancient scholar Pāṇini, likely between the 6th and 4th century BCE. It is considered the world's first formal system of linguistics and is a primary text of the Vyākaraṇa (grammar) branch of the Vedangas. **Hence, Pair 3 is not correctly matched.**

- Q 54. B** **Both SAHI and BODH initiatives mark a significant step in India's journey towards building a robust, responsible, and globally competitive health AI ecosystem.**

Strategy for Artificial Intelligence in Healthcare for India (SAHI)

- It is the national framework to guide the responsible integration of AI into India's health system.
- Vision: To enable the safe, ethical, evidence-based, and inclusive adoption of AI across India's healthcare system.
- It leverages digital public infrastructure to expand access, improve outcomes, and establish India as a global leader in responsible healthcare AI.

- It supports ecosystem stakeholders by providing shared principles, strategic direction, etc.

Benchmarking Open Data Platform for Health AI (BODH) initiative

- Developed by: IIT Kanpur in collaboration with National Health Authority (NHA).
- Purpose: It is a privacy-preserving benchmarking platform that enables rigorous evaluation of AI models using diverse, real-world health data without sharing underlying datasets.
- It has been positioned as a digital public good under the Ayushman Bharat Digital Mission.

Other Initiatives for AI in Healthcare

- Centres of Excellence for Artificial Intelligence: AIIMS Delhi, PGIMER Chandigarh, and AIIMS Rishikesh lead the development of indigenous AI solutions.
- Ethical Oversight: All AI deployment follows the ICMR Ethical Guidelines (2023)
- IndiaAI Application Development Initiative under IndiaAI Mission: Aims to develop, scale, and promote the adoption of impactful AI solutions designed to tackle significant national challenges.

Hence, option (b) is the correct answer.

- Q 55. A** Under the editorship of Harish Chandra Mukherjee, the "Hindu Patriot" (also spelled Hindoo Patriot) became the primary voice of the Indigo Revolt (1859–60). **He fearlessly exposed the brutal atrocities of European indigo planters through detailed reports and fiery editorials, ultimately forcing the British government to appoint the Indigo Commission in 1860.**

G. Subramania Aiyar: He was the founder and editor of The Hindu (in Madras/Chennai) and Swadesamitran.

Dadabhai Naoroji: Known as the "Grand Old Man of India," he was associated with the newspaper Rast Goftar and focused largely on economic critiques of British rule, such as the "Drain of Wealth" theory.

Surendranath Banerjee: He was the editor of The Bengalee and a key leader in the movement against the Partition of Bengal in 1905.

Hence, option (a) is the correct answer.

- Q 56. A** • **During the Mughal empire in 17th century khud-kashta and pahi-kashta refer to peasants. Khud-kashta were residents of the village in which they held their lands. Pahi-kashta were non-resident cultivators who belonged to some other village, but cultivated lands elsewhere on a contractual basis.** People became pahi-kashta either out of choice – for example, when terms of revenue in a distant village were more favourable – or out of compulsion – for example, forced by economic distress after a famine.

- Q 57. C** • Sikhism, the youngest and the fifth largest world religion, was founded about 500 years ago by Guru Nanak in the Punjab district (of what is now India and Pakistan). Sikhism is based on Guru Nanak's teachings and those of the nine Sikh Gurus who followed him. There are about twenty million Sikhs in the world, and most of them live in India. It emphasises the belief in one Supreme Being, the Creator. It offers a simple, straight path to eternal bliss and spreads a message of love and universal brotherhood. **Hence, statement 1 is correct.**

- The Sikh faith is a distinct religion based on the teachings of the 10 Gurus, the first of whom was Guru Nanak Devji. In 1708, the tenth and the last human Sikh Guru, Guru Gobind Singhji, vested spiritual authority in the Holy Sikh Scriptures (Guru Granth Sahibji) as he declared that the line of personal/human Gurus to an end, and installed the Granth Sahib as the Guru, the symbolic representation of the ten Gurus.
- Guru Nanak famously proclaimed, "There is no Hindu, there is no Muslim," and rejected the exclusive authority and ritualism associated with the Vedas and the Quran, advocating instead for direct devotion to one formless God. **Hence, statement 2 is correct.**
- Guru Nanak established the institution of Sangat (holy congregation/community), where disciples gathered regardless of caste or creed to pray and sing hymns together, forming a foundation for social equality. **Hence, statement 3 is correct.**

- Q 58. A**
- While BS-VI Phase 2 (introduced in 2023) already incorporated Real Driving Emissions (RDE), BS-VII is expected to expand this testing more comprehensively. This includes wider testing conditions (ambient temperature, altitude) and the potential introduction of On-Board Monitoring (OBM), which tracks emissions continuously in real-time rather than just detecting malfunctions. **Hence, statement 1 is correct.**
 - Each successive stage of BS norms is designed to be more stringent. BS-VII aims to further reduce Nitrogen Oxides (NOx) and Particulate Matter (PM). For instance, it is expected to harmonize emission limits across fuel types (gasoline and diesel), forcing diesel vehicles to meet the much lower NOx limits currently set for petrol vehicles. **Hence, statement 2 is correct.**
 - BS norms are technology-forcing. While they require high-quality fuel (like the 10 ppm sulfur fuel introduced for BS-VI), they also necessitate significant changes in vehicle engines and emission control technologies (e.g., advanced catalytic converters, DPF, and SCR systems) to meet the stricter tailpipe limits. **Hence, statement 3 is not correct.**
 - Historically, Bharat Stage norms have been modeled after European standards. BS-VI was aligned with Euro 6. Therefore, BS-VII is expected to align with the Euro 7 standards, which include stricter limits and, for the first time, regulate non-exhaust emissions like brake and tire wear particles. **Hence, statement 4 is correct.**

- Q 59. B**
- The Deccan Riots of 1875 were primarily an agrarian uprising in the Poona and Ahmednagar districts of Maharashtra. The movement was driven by specific economic factors:
- **Oppressive practices of moneylenders: Moneylenders (Sahukars) exploited peasants through exorbitant interest rates, manipulation of account books, and using the British legal system to seize peasant lands for debt recovery. Hence, point 1 is correct.**
 - **Impact of the American Civil War: During the American Civil War (1861–65), Britain turned to India for cotton, leading to a boom in prices and credit. After the war ended in 1865, cotton prices crashed, and credit dried up. This collapse, combined with a high British land revenue hike, pushed the already indebted peasants to the brink of rebellion. Hence, point 3 is correct.**
 - **Exorbitant Interest Rates: Desperate peasants were exploited by moneylenders through exorbitant interest rates on loans. The high interest often made it impossible for peasants to repay the principal, trapping them in a cycle of endless indebtedness.**

- **Land Revenue Policies:** The British colonial administration's rigid land revenue system, particularly the Ryotwari settlement, imposed direct taxes on peasants based on their land holdings.
- **Legal Bias:** The colonial legal system was heavily skewed in favour of moneylenders. Courts typically sided with moneylenders in disputes over debts and land, leaving peasants with little recourse in legal matters.

Interference of the British in their religious practices: While religious interference was a major grievance during the Revolt of 1857, it was not a significant reason for the Deccan Riots of 1875. The Deccan Riots were strictly an economic and social conflict between the ryots (peasants) and the moneylenders/colonial revenue administration. **Hence, point 2 is not correct.**

- Q 60. C**
- **Establishment of the Bethune School (1849):** Originally founded as the Hindu Female School by John Elliot Drinkwater Bethune in Calcutta, it was the first fruit of the effort to provide secular education to women in India.
 - **Wood's Despatch (1854):** Known as the "Magna Carta of English Education in India," this formal despatch by Sir Charles Wood outlined a comprehensive plan for spreading education, including the establishment of departments of education in provinces and universities in presidency towns.
 - **Establishment of the Aligarh Muhammadan Anglo-Oriental College (1875):** Founded by Sir Syed Ahmed Khan, this institution was central to the Aligarh Movement and later became the Aligarh Muslim University (AMU).
 - **Hunter Commission on Education (1882):** Appointed by Lord Ripon and presided over by Sir William Hunter, this commission was tasked with reviewing the progress of education since Wood's Despatch, focusing particularly on primary and secondary education.
 - **Hence, option (c) is the correct answer.**

- Q 61. C**
- **The constitution of the National Company Law Tribunal (NCLT) and National Company Law Appellate Tribunal (NCLAT) was done in 2016 under the companies Act, 2013.** These bodies have been constituted for faster resolution of corporate disputes and to reduce the multiplicity of agencies thereby promoting 'ease of doing business' in the country. **Hence statement 1 is correct.**
 - **With the constitution of the National Company Law Tribunal (NCLT), the Company Law Board (CLB) stands dissolved and cases pending with CLB were transferred to NCLT.**
 - The Insolvency and Bankruptcy Code, 2016, and Sick Industrial Companies Act (SICA) Repeal Act, 2003, were also put in place in 2016. With this, the Board of Industrial and Financial Reconstruction (BIFR) and Appellate Authority for Industrial and Financial Reconstruction were dissolved and Benches of NCLT were designated to exercise the jurisdiction, powers and authority of Adjudicating Authority conferred by or under part II of the Insolvency and Bankruptcy Code, 2016.
 - **NCLAT hears appeals against the orders of National Company Law Tribunal(s) (NCLT). It is also the Appellate Tribunal for hearing appeals against the orders passed by Insolvency and Bankruptcy Board of India. The Competition Appellate Tribunal under the provisions of Competition Act was dissolved in 2017 and merged with NCLAT. The appeal against the orders of Competition Commission of India is now with NCLAT. Hence statement 2 is correct.**

- **Central government has also notified provision for transfer of proceedings relating to arbitration, compromise, arrangements and reconstruction to the benches of NCLT.** The applications pending before High Courts on winding up shall continue to be dealt by the respective High Courts and fresh applications are required to be filed before NCLT.
- **NCLAT has its headquarters at New Delhi. At present, the NCLT has its Benches at 10 locations,** viz., New Delhi, Ahmedabad, Allahabad, Bengaluru, Chandigarh, Chennai, Guwahati, Hyderabad, Kolkata and Mumbai.

Q 62. C • Indian Association of Calcutta

- The Indian Association of Calcutta (also known as the Indian National Association) superseded the Indian League and **was founded in 1876 by younger nationalists of Bengal led by Surendranath Banerjee and Ananda Mohan Bose**, who were getting discontented with the conservative and pro-landlord policies of the British Indian Association.
- The Indian Association was the most important of pre- Congress associations and aimed to “promote by every legitimate means the political, intellectual and material advancement of the people.” It set out to:
 - **(i) create a strong public opinion on political questions; and**
 - **(ii) unify the Indian people in a common political programme.**
- It protested against the reduction of the age limit in 1877 for candidates of the Indian Civil Service examination. The association demanded simultaneous holding of civil service examination in England and India and Indianisation of higher administrative posts. **It led a campaign against the repressive Arms Act and the Vernacular Press Act.**
- **Organised the first Indian National Conference in 1883, acting as a precursor to the Indian National Congress.**
- **Hence, option (c) is the correct answer.**

- Q 63. A**
- The National Centre for Good Governance (NCGG) traces its origin to the National Institute of Administrative Research (NIAR), which was set up in 1995 by the Lal Bahadur Shastri National Academy of Administration (LBSNAA), the Government of India's topmost training institute for civil services, but it was set up in 2014. NIAR was subsequently rechristened and subsumed into NCGG. **Hence statement 1 is not correct.**
 - The National Centre for Good Governance (NCGG) was set up in 2014 by the Government of India as an apex-level autonomous institution under the Ministry of Personnel, Public Grievances and Pensions. NCGG deals with a gamut of governance issues from local, state to national levels, across all sectors. **Hence statement 2 is not correct.**
 - The Centre is mandated to work in the areas of governance, policy reforms, capacity building and training of civil servants and technocrats of India and other developing countries. It also offers training programs for civil servants from other countries as part of India's soft power diplomacy. The two-week-long 71st batch Capacity Building Programme (CBP) for the civil servants of Bangladesh organised by the National Centre for Good Governance (NCGG) in partnership with the Ministry of External Affairs (MEA) concluded on 15th March 2024. Following the completion of the first phase of CBP for 1,500 civil servants, NCGG signed an MoU with the Government of Bangladesh to enhance the capacity of an additional 1,800 civil servants by 2025. **Hence statement 3 is correct.**

- Q 64. D** • The Lingaraja Temple's origins trace to around the 7th century, with major reconstructions in the 11th century under the Somavamsi dynasty. It marks the culmination of temple-building traditions in Bhubaneswar, often called Ekamra Kshetra in sacred texts. The temple's deity, Harihara, unites aspects of Shiva and Vishnu, symbolizing syncretism in medieval Odishan worship. **Hence, Pair 1 is not correctly matched.**
- Padmanabhaswamy Temple's recorded history dates back over a millennium, with references in early Tamil literature and inscriptions. Its current form was largely completed under the Travancore royal family in the 18th century. The temple's seven-tiered gopuram (gateway tower), long pillared corridors, and mural paintings exemplify Dravidian artistry fused with Kerala's indigenous temple traditions. **Hence, Pair 2 is not correctly matched.**
 - Srikalahasti Temple, also known as Sri Kalahasteeswara Swamy Temple, is an ancient Hindu shrine in Srikalahasti town, Andhra Pradesh, India. Dedicated to Lord Shiva as Kalahasteeswara, it is one of the five Pancha Bhoota Sthalas, representing the element of air (Vāyu). Revered as "Dakshina Kailasam," it is among South India's most significant Shaivite pilgrimage sites. The earliest structure dates to the Pallava period, with major additions under the Cholas and Vijayanagara rulers. The complex displays classical Dravidian architecture—carved stone corridors, mandapams, and towering gopurams. The 120 ft Raja Gopuram built by Krishnadevaraya remains its defining landmark. **Hence, Pair 3 is not correctly matched.**
- Q 65. D** • **The Securities Transaction Tax (STT) is a type of financial institution tax levied in India on transactions done on the domestic stock exchange.** The rates of STT are prescribed by the Central / Union Government through its Budget from time to time. In tax parlance, this is categorised as a direct tax.
- **Capital Gains Tax is a direct tax applies on the sale of all assets if a profit has been made by the owner of the asset.** The tax is further classified into Short term capital gains and Long term capital gains. **Hence statement 1 is not correct.**
 - **Minimum Alternate Tax is a direct tax aimed at the taxation of 'zero tax companies' i.e., those companies which show zero or negligible income to avoid tax.** Under MAT, such companies are made liable to pay to the government, by deeming a certain percentage of their book profit as taxable income. **Hence statement 2 is not correct.**
 - **MAT is levied at the rate of 18.5% of the book profits.** MAT rate has been progressively increased from 7.5% in 2000 to 18.5% in 2015.
- Q 66. A** • The Members of Parliament Local Area Development Division is entrusted with the responsibility of implementation of Members of Parliament Local Area Development Scheme (**MPLADS**). **Under the scheme, each MP has the choice to suggest to the District Collector for works to the tune of Rs.5 Crores per annum to be taken up in his/her constituency.**
- **Role of MP in MPLADS:**
 - **recommend works of developmental nature with emphasis on the creation of durable community assets based on the locally felt needs to be taken up in their Constituencies.** **Hence, option (a) is the correct answer.**
 - see the progress of works of the recommend works.

- **However, State/UT monitors and supervises** the implementation of MPLAD Scheme in in the entire State/UT.
- **Ministry of Statistics and Programme Implementation update the releases of MPLADS Funds**, monitors the progress of work through Work Management System (WMS) and monitors the expenditure at macro level.

Q 67. C • Organ Donation in India is regulated under the Transplantation of Human Organs Act of 1994

- National Network division of National Organ & Tissue Transplant Organisation (**NOTTO**) function as apex center for All India activities of coordination and networking for procurement and distribution of Organs and Tissues and registry of Organs and Tissues Donation and Transplantation in the country. **Hence, statement 3 is correct.**
- (NOTTO) is a National level organization set up under Directorate General of Health Services, Ministry of Health and Family Welfare, Government of India.
- In India, the organs(a part of the body that performs a specific function) that can be donated are: Liver, Kidney, Pancreas, Heart, Lung, Intestine. **Hence, statement 2 is not correct.** The tissues that can be donated are: Cornea, Bone, Skin, Heart Valve, blood vessels, nerves, and tendon etc.

Who can be a Donor?

- Living Donor: Any person **not less than 18 years of age**, who voluntarily authorize the removal of any of his organ and/or tissue, during his or her lifetime, as per prevalent medical practices for therapeutic purposes. **Hence, statement 1 is not correct.**
- Deceased Donor: Anyone, regardless of age, race or gender can become an organ and tissue donor after his or her Death (Brainstem/Cardiac). Consent of near relative or a person in lawful possession of the dead body is required. If the deceased donor is under the age of 18 years, then the consent required from one of the parents or any near relative authorized by the parents is essential. Medical suitability for donation is determined at the time of death.

About NOTTO: It has following two divisions:

- **National Human Organ and Tissue Removal and Storage Network:** This has been mandated as per the Transplantation of Human Organs (Amendment) Act 2011. It would function as apex centre for All India activities of coordination and networking for procurement and distribution of Organs and Tissues and registry of Organs and Tissues Donation and Transplantation in the country. The network will be established initially for Delhi and gradually expanded to include other States and Regions of the country.
- **National Biomaterial Centre(National Tissue Bank):** The Transplantation of Human Organs (Amendment) Act 2011 has included the component of tissue donation and registration of tissue Banks. It becomes imperative under the changed circumstances to establish National level Tissue Bank to fulfill the demands of tissue transplantation including activities for procurement, storage and fulfil distribution of biomaterials.

The main thrust & objective of establishing the centre is to fill up the gap between 'Demand' and 'Supply' as well as 'Quality Assurance' in the availability of various tissues.

Q 68. B • The idea of creation of Zonal Councils was mooted by the first Prime Minister of India, Pandit Jawahar Lal Nehru in 1956. In the light of the vision of Pandit Nehru, five Zonal Councils were set

up vide Part-III of the States Re-organisation Act, 1956. The present composition of each of these Zonal Councils is as under:

- The Northern Zonal Council, comprising the States of Haryana, Himachal Pradesh, Jammu & Kashmir, Punjab, Rajasthan, National Capital Territory of Delhi and Union Territory of Chandigarh;
- **The Central Zonal Council, comprising the States of Chhattisgarh, Uttarakhand, Uttar Pradesh and Madhya Pradesh. Hence, option (b) is the correct answer.**
- The Eastern Zonal Council, comprising the States of Bihar, Jharkhand, Orissa, Sikkim and West Bengal;
- The Western Zonal Council, comprising the States of Goa, Gujarat, Maharashtra and the Union Territories of Daman & Diu and Dadra & Nagar Haveli;
- The Southern Zonal Council, comprising the States of Andhra Pradesh, Karnataka, Kerala, Tamil Nadu and the Union Territory of Puducherry.
- The North Eastern States i.e. (i) Assam (ii) Arunachal Pradesh (iii) Manipur (iv) Tripura (v) Mizoram (vi) Meghalaya and (vii) Nagaland are not included in the Zonal Councils and their special problems are looked after by the North Eastern Council, set up under the North Eastern Council Act, 1972.
- **The main objectives of setting up of Zonal Councils are as under :**
 - Bringing out national integration;
 - Arresting the growth of acute State consciousness, regionalism, linguism and particularistic tendencies;
 - Enabling the Centre and the States to co-operate and exchange ideas and experiences;
 - Establishing a climate of co-operation amongst the States for successful and speedy execution of development projects.

Q 69. D AWD is a water management technique where rice fields are intermittently dried and re-flooded, rather than being kept under continuous standing water. **Hence, statement 1 is correct.**

Methane is produced in flooded, anaerobic (oxygen-poor) soils. By periodically drying the field, oxygen is introduced into the soil, which inhibits methane-producing bacteria and significantly reduces emissions. **Hence, statement 2 is correct.**

While AWD reduces methane, the alternating wet and dry cycles (nitrification and denitrification processes) can lead to an increase in nitrous oxide emissions. Research often highlights this "trade-off," though the net global warming potential usually remains lower than continuous flooding. **Hence, statement 3 is correct.**

By reducing the time the field is flooded, AWD can reduce water use by up to 25–30% without significantly impacting crop yields. **Hence, statement 4 is correct.**

Q 70. C Global Biofuel Alliance (GBA): This was an India-led initiative launched by Prime Minister Narendra Modi on September 9, 2023, during the G20 Summit in New Delhi. India is one of the founding members along with the USA and Brazil. **Hence, point 1 is correct.**

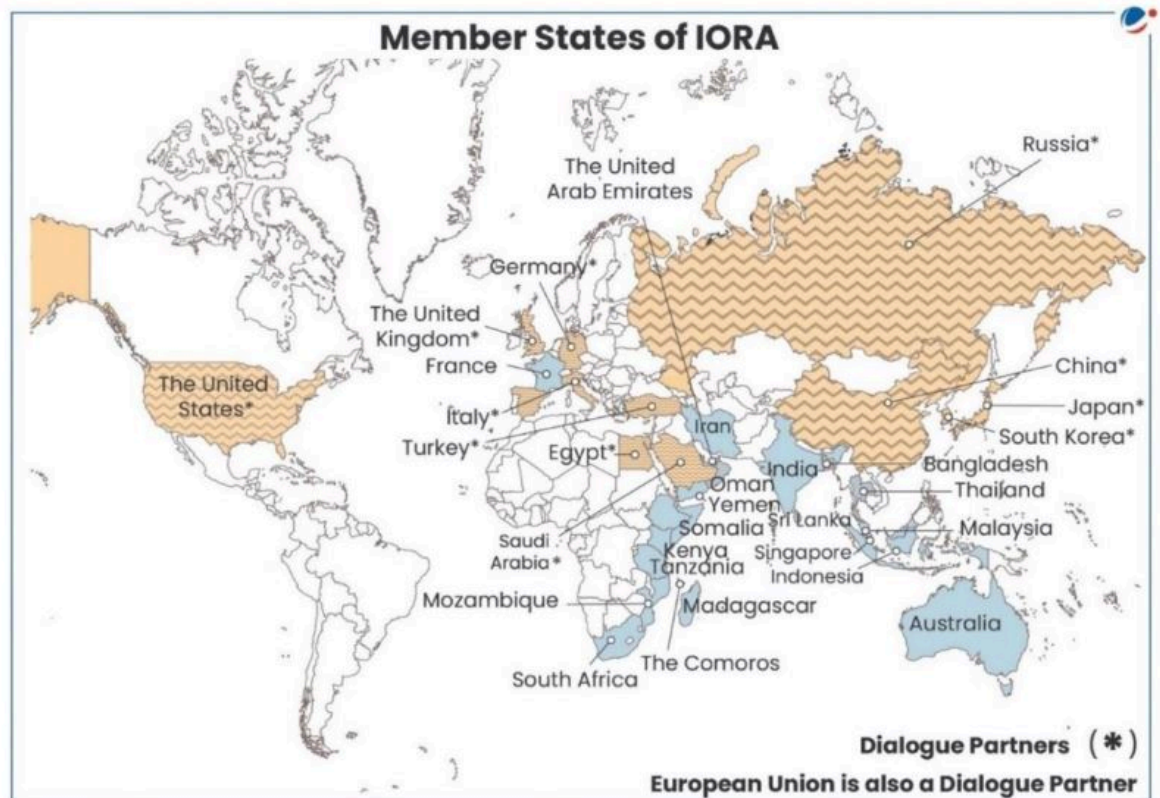
Global Methane Pledge: India has not joined this initiative. It was launched at COP26 in 2021 by the US and EU to reduce global methane emissions by 30% by 2030. India declined to sign because methane emissions in the country are primarily from agriculture (survival emissions for small farmers) rather than industrial sectors. **Hence, point 2 is not correct.**

Mangrove Alliance for Climate (MAC): India joined this alliance as a founding member during the COP27 summit in Egypt in 2022. The initiative was led by the UAE and Indonesia. **Hence, point 3 is correct.**

International Solar Alliance (ISA): This was co-founded by India and France. It was launched in 2015 during the COP21 summit in Paris. Its headquarters (Secretariat) is located in Gurugram, India. **Hence, point 4 is correct.**

Hence, option (c) is the correct answer.

Q 71. D •



- The Indian Ocean Rim Association (IORA), established in 1997, is a leading intergovernmental organization comprising 23 member states and 12 dialogue partners bordering the Indian Ocean. Headquartered in Ebène, Mauritius, it fosters regional economic cooperation, sustainable development, and maritime security. **India, a key driver, will assume the chair for 2025-2027.**
- **Thus, option (d) is the correct answer.**

Q 72. A •

- Bajji Rao I (1720-1740) was the eldest son of Balaji Viswanath. He succeeded his father as Peshwa at the age young age of twenty. The Maratha power reached its zenith under him.
- He initiated the system of confederacy among the Maratha chiefs. Under this system, each Maratha chief was assigned a territory which could be administered autonomously. As a result, many Maratha families became prominent and established their authority in different parts of India. They were the Gaekwad at Baroda, the Bhonsle at Nagpur, the Holkars at Indore, the Scindias at Gwalior, and the Peshwas at Poona. **Hence, statement 1 is correct.**

- Ahmad Shah Abdali invaded India during the Peshwaship of Balaji Baji Rao (1740-1761). **Hence, statement 2 is not correct.**

Q 73. D East Asia Summit (EAS):

- **The East Asia Summit (EAS) is an annual premier leader-led forum for strategic dialogue on political, security, and economic issues in the Indo-Pacific, established in 2005 . It promotes regional peace and prosperity through ASEAN-centered, top-level diplomatic engagement**
- Established: 2005, Kuala Lumpur, Malaysia
- Nature: Leaders-led forum for dialogue on strategic, political, and economic issues.
- **Members: 18 Members: ASEAN + Australia, China, Japan, India, New Zealand, South Korea, Russia, US. Hence, option (d) is the correct answer.**

Q 74. A • The World Economic Outlook (WEO) is published by the International Monetary Fund (IMF) twice a year. As per this report, India has to grow at a rate of 8% at constant prices on average for a decade or two in order to realise its economic aspirations of becoming Viksit Bharat by 2047. Apart from this, India also needs to boost investment to 35% of GDP, create 7.85 million new non-farm jobs annually, achieve 100% literacy, and build future ready infrastructure by 2030. **Hence pair 1 is not correctly matched.**

- The Trade Monitoring Report is published by the World Trade Organisation (WTO). It tracks trade policies, tariff changes, non-tariff measures, and protectionist trends. Recently, it highlighted concerns regarding deglobalisation trends in global trade due to geo-economic fragmentation. **Hence, pair 2 is not correctly matched.**
- The World Energy Outlook is published annually by the International Energy Agency (IEA). It provides a comprehensive analysis of global energy demand, supply trends, energy security, and renewable energy developments. As per Economic Survey 2024-25, this report highlights concerns over China's dominance in renewable energy. This may pose a challenge in case of sustainable renewable energy transitions. **Hence, pair 3 is correctly matched.**

Q 75. A • The Fact: India is endowed with large resources of iron ore, primarily Hematite and Magnetite. Hematite is considered the most important industrial iron ore due to its high metal content (60-70%).

- Distribution: Over 79% of hematite deposits are found in the Eastern Sector (Odisha, Jharkhand, Chhattisgarh), while magnetite is mostly concentrated in the Southern Sector (Karnataka). Overall, India's reserves and production are heavily dominated by the hematite variety. **Hence, statement 1 is correct.**
- Petroleum is not found in the Dharwar systems. The Dharwar system consists of ancient metamorphic rocks (2.5 billion years old) which are famous for metallic minerals like gold and manganese. Petroleum and natural gas are associated with Sedimentary rocks of the Tertiary period (roughly 15 to 60 million years old). Major areas include the Brahmaputra valley (Assam), Gujarat coast, and offshore basins like Mumbai High and the Krishna-Godavari (KG) basin. **Hence, statement 2 is not correct.**
- Gondwana Coal: It is much older (approx. 250 million years) and constitutes 98% of India's coal reserves. It is largely Bituminous coal, which has a higher carbon content (60% to 90%) and lower moisture.

- Tertiary Coal: It is much younger (15 to 60 million years old). It is generally Lignite (brown coal), which has a lower carbon content and higher moisture/sulphur content.
- **Hence, statement 3 is not correct.**

- Q 76. A**
- **Article 348(1)(a)** of the Constitution of India states that all proceedings in the Supreme Court and in every High Court, shall be in English language. The authoritative texts of all Bills to be introduced or amendments thereto to be moved in either House of Parliament or in the House or either House of the Legislature of a State, of all Acts passed by Parliament or the Legislature of a State and of all Ordinances promulgated by the President or the Governor of a State, and of all orders, rules, regulations and bye-laws issued under this Constitution or under any law made by Parliament or the Legislature of a State, shall be in English language.
 - **Clause (2) of the Article 348** of the Constitution states that notwithstanding anything in sub-clause (a) of clause (1), **the Governor of a State may, with the previous consent of the President, authorize the use of Hindi Language, or any other language used for any official purposes of the State, in proceedings in the High Court having its principal seat in that State. Hence, statement 1 is correct.**
 - **Under Article 235** of the Constitution of India, the administrative control over the district and subordinate judiciary in the States vest with the concerned High Court. **As regards, use of Hindi or regional language in lower courts, it is decided by the High Court and State Government concerned in consultation with each other. Hence, statement 2 is not correct.**

- Q 77. B**
- **Under Article 112 of the Constitution**, a statement of estimated receipts and expenditure of the Government of India has to be laid before Parliament in respect of every financial year. The Annual Financial Statement shows the receipts and payments of Government under the three parts in which Government accounts are kept: (i) Consolidated Fund, (ii) Contingency Fund and (iii) Public Account
 - **The following expenditure shall be expenditure charged on the Consolidated Fund of India-**
 - (a) the emoluments and allowances of the President and other expenditure relating to his office;
 - (b) the salaries and allowances of the Chairman and the Deputy Chairman of the Council of States and the Speaker and the Deputy Speaker of the House of the People;
 - (c) debt charges for which the Government of India is liable including interest, sinking fund charges and redemption charges, and other expenditure relating to the raising of loans and the service and redemption of debt;
 - (d) (i) the salaries, allowances and pensions payable to or in respect of Judges of the Supreme Court;
 - (ii) the pensions payable to or in respect of Judges of the Federal Court;
 - (iii) the pensions payable to or in respect of Judges of any High Court which exercises jurisdiction in relation to any area included in the territory of India or which at any time before the commencement of this Constitution exercised jurisdiction in relation to any area included in a Governor's Province of the Dominion of India;
 - (e) the salary, allowances and pension payable to or in respect of the Comptroller and Auditor-General of India;

- (f) any sums required to satisfy any judgment, decree or award of any court or arbitral tribunal; Hence, statement 2 is correct.
- (g) any other expenditure declared by this Constitution or by Parliament by law to be so charged.
- **Article 113 of the Constitution, states that expenditure charged upon the Consolidated Fund of India shall not be submitted to the vote of Parliament but can be discussed in either House of Parliament of any of those estimates. Hence, statement 1 is not correct.**

Q 78. B • **DRSCs were set up in 1993 to scrutinize the functioning of the various Ministries/ Departments of the Union Government** assigned to them in order to further strengthen the accountability of the Government to Parliament.

- **Twenty-four DRSCs have been constituted consisting of not more than thirty-one members,** out of which twenty-one members are nominated by the Speaker, Lok Sabha and ten members are nominated by the Chairman, Rajya Sabha. **Hence, statement 1 is correct.**
- **These Committees are entrusted with the following functions:**

I to consider the Demands for Grants of the related Ministries/ Departments and report thereon;

II to examine Bills, pertaining to the related Ministries/Departments, referred to the Committee and report thereon;

III to consider the annual reports of the Ministries/ Departments and report thereon; and

IV to consider national basic long term policy documents and report thereon. Hence, statement 2 is correct.

- **Eight DRSCs function under the control and direction of the Chairman, Rajya Sabha, while sixteen such Committees function under the control and direction of the Speaker, Lok Sabha. Hence, statement 3 is not correct.**

Q 79. B • **Nagorno-Karabakh Region is a disputed territory between Armenia and Azerbaijan.** A mountainous region located inside Azerbaijan's internationally recognised borders. Historically, it had a majority ethnic Armenian population, which sought self-rule.

- **Musa island is claimed by both UAE and Iran.** Abu Musa is a 12.8-square-kilometre island in the eastern Persian Gulf, near the strategic Strait of Hormuz, currently administered by Iran as part of Hormozgan province. It is central to a long-standing territorial dispute with the United Arab Emirates (UAE), which claims the island, while Iran has maintained control since 1971
- **Senkaku Island is a disputed island between Japan and China.** A group of uninhabited islands in the East China Sea. Japan administers them, calling them the Senkaku Islands. China claims them, calling them Diaoyu Islands.

- **Hence, option (b) is the correct answer.**

Q 80. B • **The Mahavamsa was composed much later, around the 5th or 6th century CE (likely during the reign of King Dhatusena of Anuradhapura).** Mahavamsa, "The Great Chronicle" otherwise known as "The Great Dynasty" is the single most important work of Lankan origin (written in Pali language). **Hence, statement 1 is not correct.**

- **It describes the life and times of the people who forged our nation, from the coming of Vijaya in 543 BCE to the reign of King Mahasena (334 – 361) (6th Century BC to 4th Century AD).** A

companion volume, the Culavamsa or Choolavansha ("lesser chronicle"), covers the period from the 4th century to the British takeover of Sri Lanka in 1815.

- The Mahavamsa itself is actually comprised of three parts, all written at different times in Lankan history. The combined work, sometimes collectively referred to as the "Mahavamsa" or "Mahawansha", provides a continuous historical record of over two millennia and can be considered as the world's longest unbroken historical record. **Hence, statement 2 is correct.**
- It has been inscribed in the UNESCO Memory of the World International Register. In 2023, the Mahavamsa (specifically the version covering the period from the 6th century BCE to 1815 CE) was officially included in the UNESCO Memory of the World International Register to recognize its global significance as a continuous historical record. **Hence, statement 3 is correct.**

Q 81. A During the normal operation of coal-based thermal power plants, the combustion of coal releases several major pollutants into the atmosphere, along with solid waste products. The key pollutants include:

- Carbon Dioxide: A major greenhouse gas produced in large quantities by burning carbon-based coal.
- Sulfur Dioxide: Formed from the sulfur content in coal; it causes acid rain and respiratory illnesses.
- Nitrogen Oxides: Created from the combustion of nitrogen in the fuel and air, contributing to smog and respiratory problems. **Hence, point 4 is correct.**
- Particulate Matter (PM10 and PM2.5): Fine ash and soot particles released into the air, which can penetrate deep into the lungs.
- Mercury and Trace Metals: Toxic heavy metals like mercury, arsenic, and lead that are released, posing serious health risks. **Hence, point 2 and 3 are correct.**
- Fly Ash and Bottom Ash: Solid waste residues, which, if not managed properly, can contaminate soil and water (sometimes called fugitive emissions if they become airborne). **Hence, point 1 is correct.**
- Carbon Monoxide: Produced due to incomplete combustion.

Methane: While methane is a potent greenhouse gas associated with coal, it is primarily released during the mining and extraction process (known as coalbed methane). **Hence, point 5 is not correct.**

Hence, option (a) is the correct answer.

Q 82. B Defense minister flagged-off Akash 3rd and 4th Regiment Combat systems and inaugurated Mountain Fire Control Radar at Bharat Electronics Limited (BEL) in Bengaluru.

Akash is an indigenously developed short-range surface-to-air missile (SAM). It was designed by the Defence Research and Development Organisation (DRDO) as part of the Integrated Guided Missile Development Programme (IGMDP). **Hence, statement 1 is correct.**

The missile is powered by an Integrated Ramjet Rocket Engine. Specifically, it uses a solid-fuel ramjet propulsion system that allows it to maintain supersonic speeds (Mach 2.5) throughout its flight without deceleration. **Hence, statement 2 is correct.**

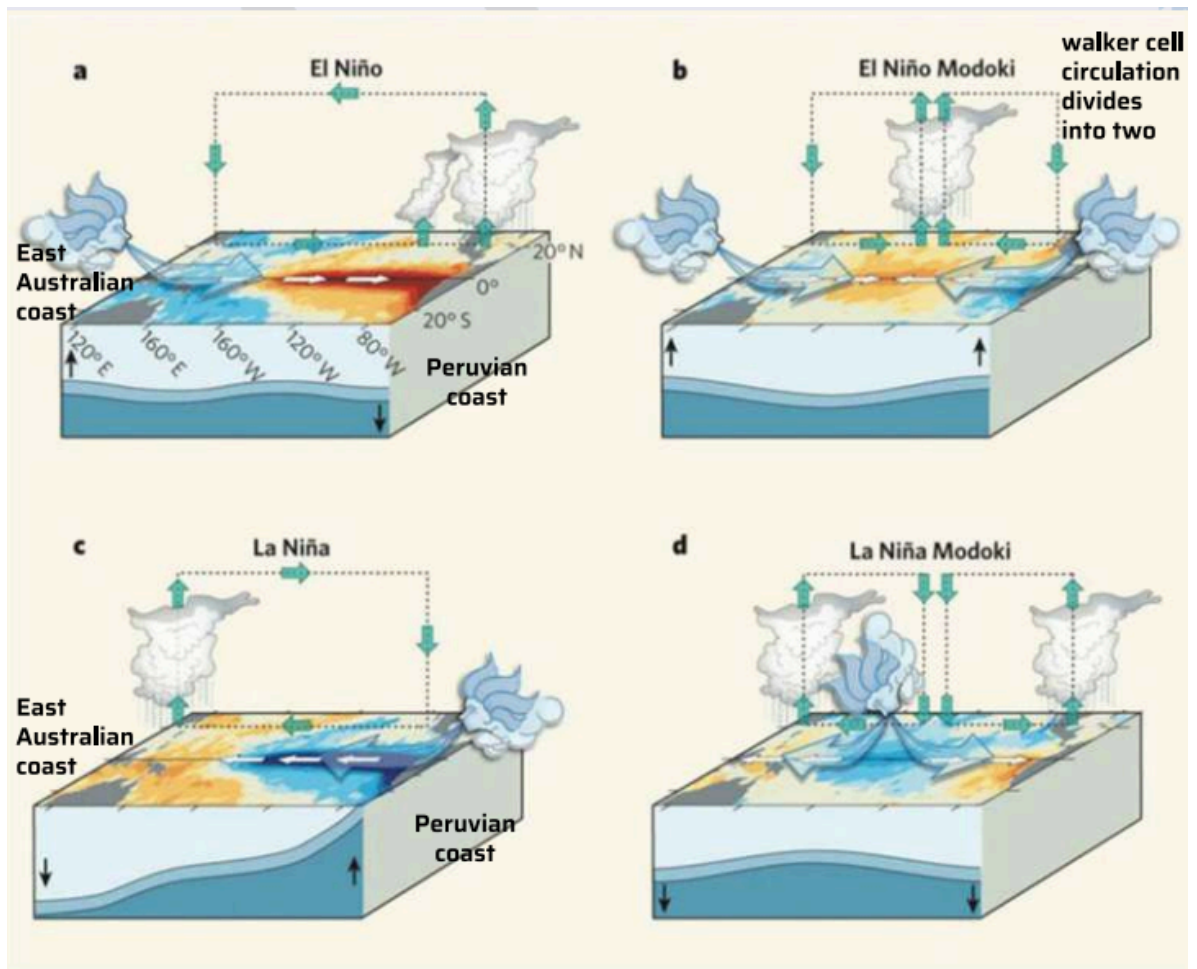
The Akash missile system, particularly its advanced variants (Akash-1S, Akash Prime, and Akash-NG), uses an indigenous Active Radio Frequency (RF) Seeker for terminal guidance, not an infrared (IR) seeker. **Hence, statement 3 is not correct.**

Akash is a critical component of India's multi-layered air defence architecture. It is designed to protect strategic assets and vulnerable areas from various aerial threats, including fighter jets, cruise missiles, and drones. It operates alongside other systems like the S-400 (long-range) and BARAK-8 to provide comprehensive coverage.

- Q 83. D**
- A Special Category Status (SCS) is a classification given by the Centre to assist in the development of those states that face geographical and socio-economic disadvantages like hilly terrains, strategic international borders, economic and infrastructural backwardness, and non-viable state finances. The concept of a special category status was first introduced in 1969 when the fifth Finance Commission sought to provide certain disadvantaged states with preferential treatment in the form of central assistance and tax breaks, establishing special development boards, reservation in local government jobs, educational institutions, etc. The Constitution of India does not include any provision for the categorization of any state in India as a 'special category state'. **Hence statement 1 is incorrect.**
 - The decision to grant special category status lies with the National Development Council, composed of the prime minister, union ministers, chief ministers and members of the planning commission, who guide and review the work of the commission. Special category status for plan assistance has been granted in the past by the National Development Council (NDC) to some states that are characterized by a number of features necessitating special consideration. **Hence statement 2 is incorrect.**
 - In November 22, the Chief Minister Nitish Kumar-led Cabinet passed a resolution seeking the grant of special category status (SCS) to Bihar. The demand comes in the backdrop of the findings from the "Bihar Caste-based Survey, 2022", which revealed that nearly one-third of Bihar's population continues to live in poverty. **Hence statement 3 is incorrect.**

Hence option (d) is the correct answer.

- Q 84. B**
- The description provided in Statement 1 actually defines El Niño Modoki, not La Niña Modoki.
 - **La Niña Modoki:** Is a phenomenon where there is anomalous cooling in the central Pacific, while the eastern and western Pacific regions experience warming.
 - Conventional La Niña: Is characterized by cooling primarily in the eastern Pacific. **Hence, statement 1 is not correct.**
 - **El Niño:** Generally suppresses cyclone activity in the Bay of Bengal due to high vertical wind shear (strong winds at different altitudes that "tear" a developing cyclone apart).
 - La Niña (and especially La Niña Modoki): The atmospheric conditions shift. In a La Niña Modoki year, the cooling in the central Pacific creates a specific pressure gradient that reduces wind shear and increases moisture over the Bay of Bengal.
 - Consequently, La Niña Modoki years typically see more frequent and intense cyclogenesis in the Bay of Bengal compared to El Niño years (which are usually 'quiet' for cyclones). **Hence, statement 2 is correct.**
 -



- Q 85. C
- Secured Overnight Financing Rate (SOFR) is a benchmark interest rate for derivatives and loans in USD that has replaced the London Interbank Offered Rate (LIBOR). **Hence statement 1 is correct.**
 - SOFR is like a 'price tag' for how much banks pay to borrow money from each other on a very short basis (overnight), using U.S. Treasury bonds as collateral.
 - The transition from LIBOR to SOFR in June 2023 marked a significant shift in interest rate benchmarks, enhancing reliability in financial markets.
 - The Secured Overnight Financing Rate (SOFR) changes daily, as it's published by the Federal Reserve Bank of New York each business day at 8 AM ET. The SOFR is based on treasury market movements, so it changes whenever the market is open. **Hence statement 2 is correct**

- Q 86. C
- Organisms that obtain their nutrition from decaying and dead organic matter are called Saprophytes (also known as saprotrophs or saprobes). These organisms perform extracellular digestion by secreting digestive enzymes onto the dead matter to break down complex organic compounds into simpler, soluble substances, which they then absorb through their cell membranes. Common examples include:
 - Fungi: Such as mushrooms, bread mold, and yeast.
 - Bacteria: Many types of soil bacteria. **Hence, option (c) is the correct answer.**
 - Other options:**
 - (a) Sporophyte: This refers to a specific multicellular stage in the life cycle of a plant or alga that produces spores. It is an anatomical phase of development rather than a mode of nutrition.

- (b) Parasite: These organisms live on or inside a living host and obtain nutrients directly from the host's body, often causing it harm.
- (d) Endophyte: This is an organism, often a fungus or bacterium, that lives inside a plant for at least part of its life cycle without causing apparent disease. They typically have a symbiotic or mutualistic relationship with the living plant rather than feeding on dead matter.

Q 87. A "Miyawaki" has become a ubiquitous buzzword in recent years, especially in urban greening, reforestation, and environmental restoration. The term has gained widespread popularity because of its association with creating dense, and "so-called" biodiverse forests with native trees in a relatively short period.

The Miyawaki technique strictly emphasises the use of native (indigenous) species. It explicitly avoids exotic or invasive species to ensure the forest is resilient and supports local biodiversity. **Hence, statement 1 is not correct.**

It involves planting a diverse mix of native species (trees, shrubs, and canopy layers) very densely (often 3–4 saplings per square meter) to mimic a mature natural forest. **Hence, statement 2 is correct.**

Because the plants are packed so closely, they compete for sunlight, which triggers rapid vertical growth. **Miyawaki forests are known to grow up to 10 times faster than conventional plantations. Hence, statement 3 is correct.**

While popularized in tropical regions like India, the technique was actually pioneered in Japan (a temperate region) and has been successfully applied in various climates globally, including Mediterranean and temperate zones. **Hence, statement 4 is not correct.**

Q 88. D 1. Temperature and Origin

- The Gulf Stream is a powerful, warm western boundary current that originates in the Gulf of Mexico.
- It carries warm, tropical water toward the North Atlantic.

2. Geographic Path

- It flows along the eastern coast of North America (the U.S. Eastern Seaboard) before turning eastward at the Grand Banks near Newfoundland.
- While the Florida Current is also warm and flows along North America, it is specifically the "upstream" portion that feeds into the much larger Gulf Stream system described in these global climate terms.

3. Climate Moderation

- As it crosses the ocean, it becomes the North Atlantic Drift. This current releases heat into the atmosphere over Western Europe.
- This process keeps European winters significantly milder (often by 10°C or more) than regions at similar latitudes in North America, such as Labrador or Quebec.

• **Hence, option (d) is the correct answer.**

• other options:

- (a) Labrador Current: This is a cold current that flows south from the Arctic along the eastern coast of Canada.
- (b) Kuroshio Current: Although it is a warm western boundary current, it is located in the Pacific Ocean, flowing along the eastern coast of Japan.

- (c) North Atlantic Equatorial Current: This current flows westward across the Atlantic near the equator; it does not flow along the North American coast or directly moderate Europe.



Q 89. D Although water covers about 71% of the Earth's surface, the vast majority is salt water in the oceans. Freshwater makes up only about 2.5% to 3% of the total water on Earth. This freshwater is distributed as follows:

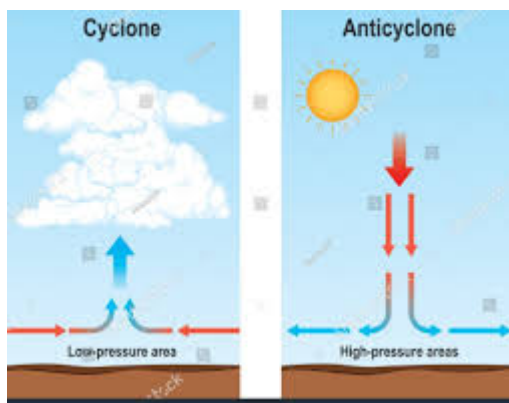
- Glaciers and Ice Caps (~68.7%): Most of Earth's freshwater is frozen in the ice sheets of Antarctica and Greenland. This is the single largest reservoir of freshwater on the planet.
- Groundwater (~30.1%): This is the second-largest source, consisting of water stored beneath the Earth's surface in soil and rock formations (aquifers).

Surface Water (~1.2%): This category includes lakes, rivers, and swamps.

- Lakes contain significantly more water than rivers, but both represent only a tiny fraction of the world's freshwater.
- Rivers actually contain the smallest amount among the options listed (less than 0.01% of all freshwater).

Hence, option (d) is the correct answer.

- Q 90. A**
- A high pressure system, also known as an anticyclone occurs when the weather is dominated by stable conditions. Under an anticyclone air is descending, forming an area of higher pressure at the surface. Because of these stable conditions, cloud formation is inhibited, so the weather is usually settled with only small amounts of cloud cover. **Hence statement 1 is correct.**
 - In the Northern Hemisphere winds blow in a clockwise direction around an anticyclone. As isobars are normally widely spaced around an anticyclone, winds are often quite light.
 - Anticyclones can be identified on weather charts as an often large area of widely spaced isobars, where pressure is higher than surrounding areas. **Hence statement 3 is not correct.**
 - In winter the clear, settled conditions and light winds associated with anticyclones can lead to frost and fog. The clear skies allow heat to be lost from the surface of the earth by radiation, allowing temperatures to fall steadily overnight, leading to air or ground frosts. **Hence statement 2 is not correct.**



- Q 91. C**
- In the age of the Rig Veda differentiation based on occupations had started but this division was not very rigid. **Hence, statement 1 is correct.**
 - There are repeated mentions of slaves who were given as gifts to the priests. They were mainly women slaves employed for domestic purposes. It is clear that in Rig Vedic times slaves were not used directly in agriculture or other producing activities. **Hence, statement 2 is correct.**

Q 92. B • Central Board of Indirect Taxes & Customs (CBIC):

- The Central Board of Indirect Taxes is under the Department of Revenue, Finance Ministry, Government of India.
- The CBIC was previously known as the Central Board of Excise and Customs (CBEC).
- It engages in policy formulation of matters concerned with the levy and collection of central excise duties, customs, central goods, and services tax and IGST, administration of these taxes, prevention of smuggling, and also narcotics to the extent prescribed.
- The CBIC is the administrative authority for its subordinate organizations, including Central Excise and Central GST Commissioners, Custom Houses, and the Central Revenues Control Laboratory.
- **CBIC is a statutory body established under the Central Boards of Revenue Act, of 1963.**

- CBIC was formed in 1964 when the Central Board of Revenue was split into the Central Board of Direct Taxes (CBDT) and the Central Board of Excise and Customs, which was renamed the Central Board of Indirect Taxes and Customs in 2018.
- The Central Board of Revenue was set up under the Central Board of Revenue Act, 1924 as the apex body of the Income Tax Department.
- **Central Board of Direct Taxes (CBDT):**
 - It **is a statutory body** established as per the Central Board of Revenue Act, of 1963. **Hence, statement 1 is not correct.**
 - It is India's official financial action task force unit.
 - It **is administered by the Department of Revenue under the Ministry of Finance. Hence, statement 2 is correct.**
 - To note that originally there was a board called the Central Board of Revenue that functioned as the apex body of the Income Tax Department. The said board was set up under the Central Board of Revenue Act, of 1924, and was in charge of both direct and indirect taxes. The Central Board of Revenue got split in 1964 into two boards:
 - Central Board of Direct Taxes
 - Central Board of Excise and Customs

Q 93. B • Recently, the Ministry of Finance launched the Foreign Currency Settlement System (FCSS) at the GIFT International Financial Services Centre (IFSC).

Foreign Currency Settlement System (FCSS):

- It is a payment system authorised by the International Financial Services Centres Authority (IFSCA) under the Payment and Settlement Systems Act, 2007. **Hence, statement I is not correct.**
- It enables foreign currency transactions between IFSC Banking Units (IBUs) to be settled locally instead of routing through the traditional correspondent banking route. **Hence, statement II is correct.**
- This system will be able to settle transactions on a real-time or near real-time basis, thereby significantly reducing the settlement time. The traditional correspondent banking route involves multiple banks in the transaction chain, taking up to 36 to 48 hours for settlement.
- The system will initially settle transactions undertaken in US dollars with scope for adding other foreign currencies over time.
- Operator of the FCSS is CCIL IFSC Limited (CCIL IFSC), a subsidiary of Clearing Corporation of India Limited.

Q 94. C • Autonomous vehicles (AVs) integrate a suite of sensors to perceive their environment. LiDAR provides high-resolution 3D maps, cameras interpret visual cues like traffic signs and lane markings, and radar detects the distance and velocity of objects, especially in poor weather. **Hence, statement 1 is correct.**

- AVs rely on Artificial Intelligence (AI) and Machine Learning algorithms to process the massive amounts of data from these sensors. These algorithms are essential for tasks like object classification, behaviour prediction, and determining the optimal driving path. **Hence, statement 2 is correct.**

- While AVs can use technologies like SLAM (Simultaneous Localisation and Mapping) to navigate when signals are unavailable, they do not function "entirely independently" of satellite-based navigation. In standard operation, GPS/GNSS is a critical component used for global positioning, route planning, and time-syncing with high-definition maps. Most systems treat GPS as a primary data source for localisation, supplementing it with other sensors only when signals are blocked (e.g., in tunnels or urban canyons). **Hence, statement 3 is not correct.**
- A core function of AV navigation is obstacle detection and avoidance. The system continuously monitors for potential hazards (pedestrians, other vehicles, debris) and uses control systems to adjust the vehicle's speed (braking/acceleration) and direction (steering) in real-time. **Hence, statement 4 is correct.**

- Q 95. C** • **An initial public offering refers to the process in which a private company sells shares to the general public for the first time. Hence statement 1 is correct.**
- This enables the company to generate fundraising through selling stakes owned to institutional and retail investors.
 - **This process changes a private-owned company into one that is publicly owned. Hence statement 2 is correct.**
 - IPOs in India are **regulated by the Securities and Exchange Board of India (SEBI)**, which ensures compliance with the rules and investor protection.
 - IPOs increase the visibility and credibility of a company, marking a milestone in its growth trajectory while enhancing market liquidity and economic development.

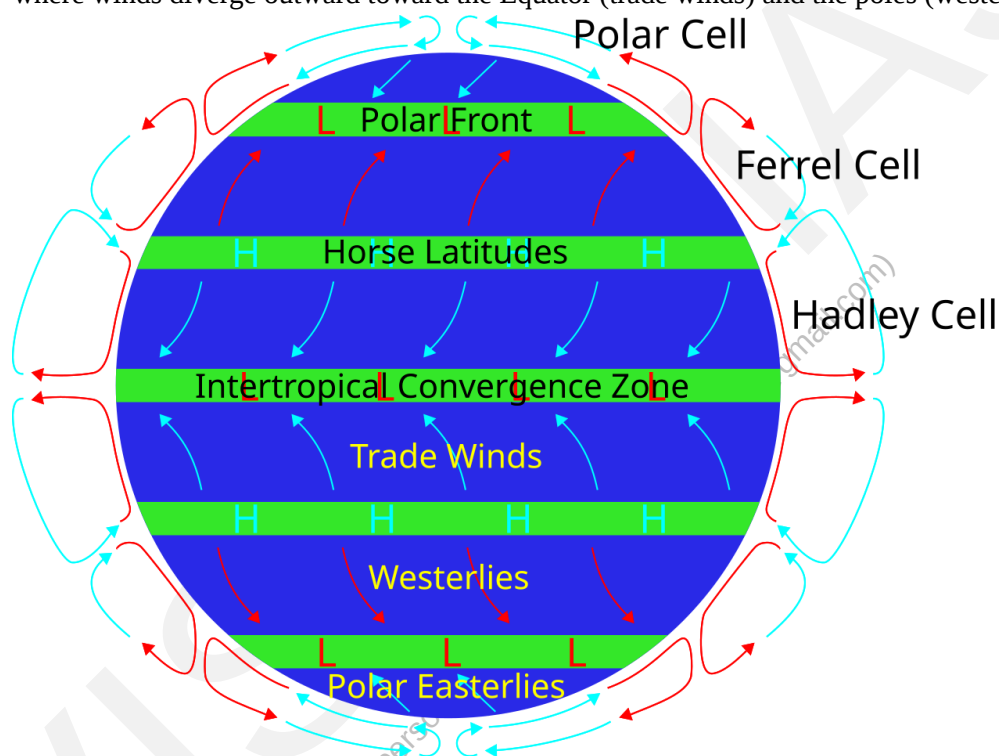
Hence option (c) is the correct answer.

Q 96. A Key Provisions of Chemical Weapons Convention (CWC):

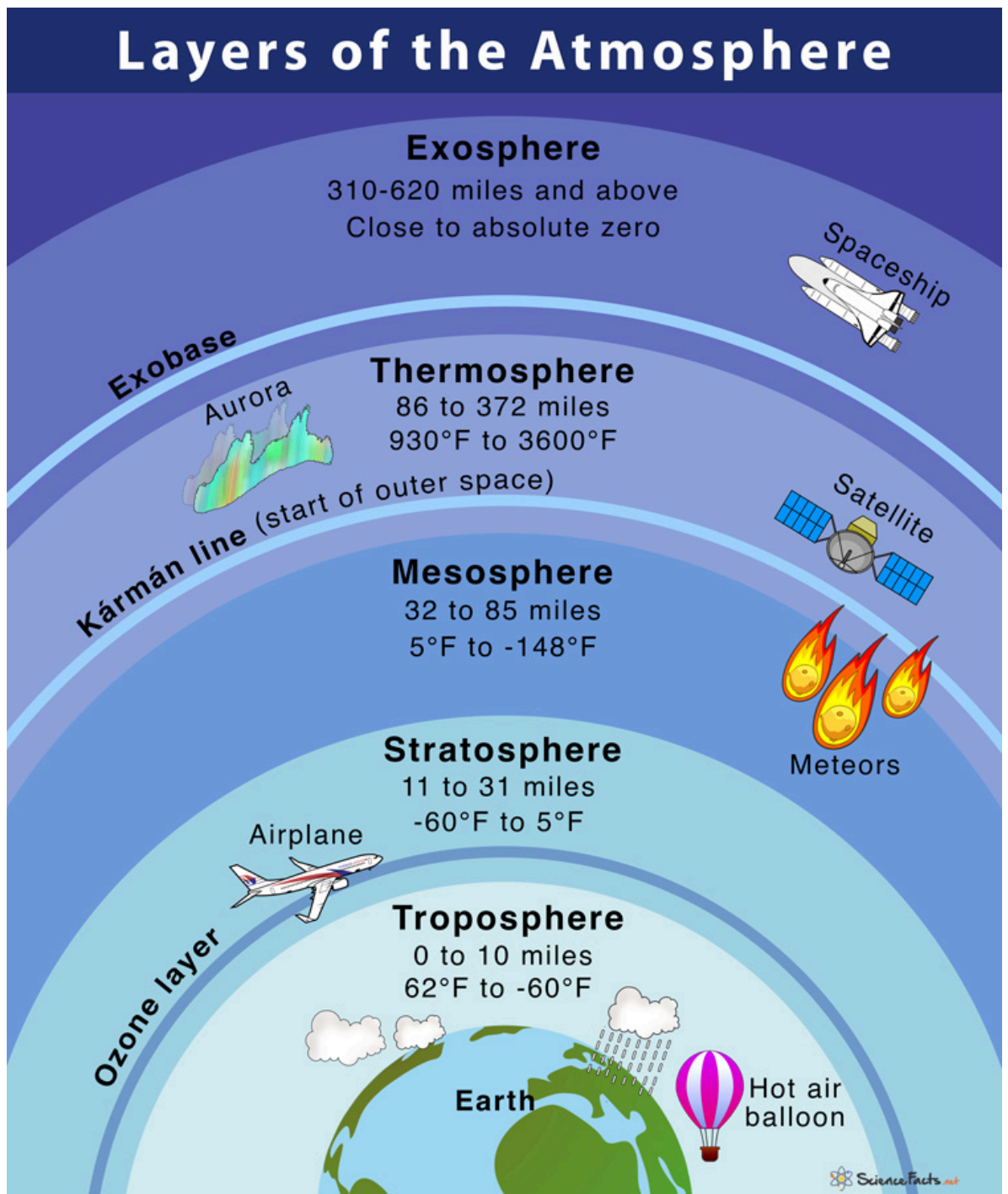
- **Destruction of Chemical Weapons: All chemical weapons stocks of member states must be destroyed within a specific timeframe.**
- **Verification:** The CWC includes detailed verification measures to ensure compliance, including inspections of chemical weapons stockpiles, facilities, and research programs.
- The Chemical Weapons Convention (CWC) is a multilateral disarmament treaty, effective since April 29, 1997, that prohibits the development, production, acquisition, stockpiling, transfer, and use of chemical weapons.
- **Non-Diversion: The Convention prohibits the diversion of chemical materials to use as weapons or for military purposes.**
- **Member states are required to adopt national laws and regulations that align with the CWC's provisions and provide for the destruction and monitoring of chemical weapons. Hence, option (a) is the correct answer.**

- Q 97. A** The Horse Latitudes (located roughly between 30° to 35° North and South) are subtropical high-pressure belts formed by the subsidence of air from the upper branches of the Hadley cell. As this air descends, it warms adiabatically, which suppresses cloud formation and results in:
- High atmospheric pressure (anticyclonic conditions).

- Calm or light, variable winds because the vertical movement of sinking air dominates over horizontal wind flow.
- Arid climates and clear skies, which is why most of the world's major deserts (like the Sahara) are found in these zones. **Hence, option (a) is the correct answer.**
- Other options:
- (b) Strong westerly winds known as the Roaring Forties: The Roaring Forties are located further poleward, between 40° and 50° South. While the Westerlies originate near the Horse Latitudes, the Horse Latitudes themselves are defined by calm, not the strong winds found in the higher latitudes.
- (c) Frequent cyclonic storms and heavy rainfall: These features describe low-pressure regions like the Equator or subpolar lows. High pressure in the Horse Latitudes actually inhibits storm development and rainfall.
- (d) Low pressure and converging trade winds: This describes the Doldrums or the Intertropical Convergence Zone (ITCZ) located near the Equator. The Horse Latitudes are high-pressure zones where winds diverge outward toward the Equator (trade winds) and the poles (westerlies)



- Q 98. A**
- In the Stratosphere, the temperature increases with altitude (temperature inversion) because the ozone layer absorbs UV radiation. The Stratopause is the transition boundary at the top of the stratosphere (approx. 50 km). At this point, the concentration of ozone thins out, and the heating effect ends. Above this boundary, in the mesosphere, the temperature begins to decrease again. **Hence, statement 1 is correct.**
 - The Normal Lapse Rate refers to the decrease in temperature with increasing altitude at a rate of approximately 6.5°C per 1,000 meters. This is a specific characteristic of the Troposphere. In the layers above (Stratosphere and Thermosphere), the temperature trend either stays constant or increases, meaning the 'normal' cooling rate does not apply. **Hence, statement 2 is correct.**
 - Auroras primarily occur in the thermosphere (specifically the ionosphere) where charged solar particles collide with atmospheric gases. While meteors do burn up in the mesosphere. **Hence, statement 3 is not correct.**



- Q 99. B** • Early Tamil literature (the Sangam texts) also mentions different categories of people living in the villages – **large landowners or Vellalar, ploughmen or uzhavar and slaves or adimai**. Hence **option (b) is the correct answer**.
- It is likely that these differences were based on differential access to land, labour, and some of the new technologies.
- Q 100. B** • Tropical grasslands (like the Savanna) generally receive between 50 to 130 cm of rainfall annually, whereas temperate grasslands (like the Prairies or Steppes) usually receive 25 to 75 cm. Tropical grasslands are often a transition zone between deserts and tropical rainforests, leading to higher rainfall totals. **Hence, statement 1 is correct**.
- Temperate grasslands are known for their Chernozem (black) soils, which are incredibly rich in organic matter due to the slow decomposition of grass roots in cooler climates. In contrast, tropical

soils are often highly leached (lateritic) due to high temperatures and heavy seasonal rains, making them less fertile. **Hence, statement 2 is correct.**

- Fire is a critical ecological driver for both. It prevents the encroachment of woody shrubs and trees (maintaining the "grass" dominance), clears dead organic matter, and recycles nutrients back into the soil, which stimulates new growth. **Hence, statement 3 is correct.**
- Tropical grasslands (Savannas) are defined by having scattered trees (like Acacia or Baobab). Temperate grasslands are often largely treeless (except along river banks) because of lower rainfall and colder winters. **Hence, statement 4 is not correct.**

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